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**CardioRef: An android base mobile reference app for
Cardiovascular Diseases CVDs**

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Abstract

Cardiovascular diseases (CVD) continue to be a primary cause of death globally, with late detection and diagnosis posing significant challenges in their management. Enhancing awareness and delivering effective education on CVD are essential steps in mitigating these challenges. To bridge this gap, the CardioRef mobile application was created as a reliable, accessible, and educational resource for CVD awareness. The main goal was to develop a user-friendly mobile app that includes an integrated dictionary of CVD terminology, complemented by animations and text-based content to facilitate engaging learning experiences. Furthermore, the app emphasizes non-pharmacological strategies for the prevention and management of CVD, providing ongoing updates from credible organizations such as the World Health Organization (WHO) and the European Society of Cardiology (ESC). A quiz feature was also added to assess users' knowledge and promote active learning. The result of this initiative is a user-centric application that not only informs individuals about CVD but also offers current information, ultimately aiding in the reduction of late detection and enhancing public health literacy regarding CVD. The app's interactive elements and regular content updates have the potential to revolutionize how individuals interact with cardiovascular health education, thereby improving the management and prevention of CVD.

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I certify that the work presented in the dissertation is my own unless referenced.

Signature Maxwell Opare

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Table of Contents

Abstract.....	i
Acknowledgements.....	ii
Table of Contents	iii
List of Tables.....	v
List of Figures	vi
1 Introduction	1
1.1 Introduction.....	1
1.2 Aims and Objectives.....	2
1.3 Project Approach.....	4
1.3 Dissertation Outline	4
2 Background.....	6
2.1 Overview of impact of Noncommunicable (chronic) disease knowledge and awareness on mortality reduction.....	6
2.2 Defining CVD and types of CVD	6
2.3 Research on CVD knowledge and awareness	8
2.4 Research on mobile apps on CVD awareness	10
2.5 CVD awareness-based mobile apps.....	12
3 Methodology	15
3.1 Scrum Software Methodology.....	15
3.2 Scrum framework and artifacts	15
4 Design.....	17
4.1 User stories	17
4.2 Project Requirements.....	18
4.3 Prototyping	19
5 Implementation	25
6 Testing and Evaluation	50
7 Conclusions.....	55
7.1 Future Work	56
References.....	58
Appendix A Personal Reflection (Compulsory)	63
A.1 Reflection on Project.....	63
A.2 Personal Reflection.....	64
Appendix B Ethics Documentation (Compulsory).....	66

B.1	Ethics Pro-forma.....	66
Appendix C	Video Demonstration (Compulsory)	72
C.1	Link to the video	72
Appendix D	Other Appendices	73
	More relevant material.....	73

List of Tables

Table 1	Table showing product backlog and story point	28
Table 2	User stories with their T-shirt Estimation	28
Table 3	Refined user stories.....	29
Table 4	Final Product Backlog with priority and Planning poker estimation	29
Table 5	Sprint Backlog for Sprint One	30
Table 6	Table showing the additional user stories added to the first sprint in the second week of development	39
Table 7	Table showing the sprint backlog for Sprint Two.....	45
Table 8	Table showing results of end-user functionality test	51

List of Figures

Figure 1	An image show scrum artifact.	16
Figure 2	Use case diagram for CardioRef.....	17
Figure 3	Sequence diagram for CVD feature of the app.....	18
Figure 4	CardioRef Home UI Figma Prototype	20
Figure 5	CardioRef CVD UI Figma Prototype.....	21
Figure 6	CardioRef Selected CVD UI Figma page Prototype.....	22
Figure 7	CardioRef Dictionary UI Figma Prototype.....	23
Figure 8	CardioRef Newsletter UI Figma Prototype.....	24
Figure 9	The architecture of the application showing the various layers	25
Figure 10	A screenshot of Android Studio Ladybug with Pixel 9 emulator	26
Figure 11	A screenshot of SQL Lite database application	27
Figure 12	Part of xml code snippet of the home page.....	31
Figure 13	The homepage layout from the design view	32
Figure 14	The application home from the emulator view	32
Figure 15	Code snippet from the main activity showing image loading with Glide and the clickable functions of the cardviews.....	33
Figure 16	A screenshot of the CreateModel Class that was used to create the types of CVDs for the adaptor to display on the CVDs page.....	34
Figure 17	A screenshot of code snippet of the adaptor created for the recyclerview on CVDs page layout.....	34
Figure 18	a picture of the layout created for individual item in the recyclerview on the CVDs page	35
Figure 19	The emulator view after joining the classes and the layout together.....	35
Figure 20	A picture of the layout for the type of CVDs page.....	36
Figure 21	A picture showing the inclusion of the layout in Fig 19 in the main layout of one of the types of CVDs.....	37
Figure 22	Emulator view of combing the files in the raw file and the layout on one of the types of CVDs page.....	37
Figure 23	Picture of Burn-Down Chart for User Story U001 and U002.....	38
Figure 24	A picture of the Terminology class created for the Dictionary objects.....	39
Figure 25	Code snippet of database handler user to create the table for the Dictionary/Terminology.....	40

Figure 26	A picture of SQL Lite application showing the data added to the Terminology table.	40
Figure 27	A screenshot showing the emulator view of the Dictionary page after development	41
Figure 28	A picture showing the Burn-Down chart for Sprint One	42
Figure 29	Emulator view of completing User story U005.....	45
Figure 30	A picture of a code snippet showing how the questions were created in the CreateModels Class.....	46
Figure 31	A screenshot of the Quiz page from the emulator view	47
Figure 32	A screenshot showing the animation and Toast message when a correct answer is chosen	48
Figure 33	A screenshot showing the animation and Toast message when a wrong answer is chosen	49
Figure 34	Burn-Down chart for Sprint Two.....	50
Figure 35	A prototype of questionnaires used along with functional end-user testing .	52
Figure 36	A pie chart showing data distribution under Appearance and Design.....	53
Figure 37	Pie chart showing the data distribution of UI Testing.....	54

1 Introduction

1.1 Introduction

Cardiovascular Diseases (CVDs) according to World Health Organization are a group of heart and blood vessel disorders. It is the leading cause of death in the world with an estimated 17.9 million deaths, representing 32% of all global death each year. CVDs are projected to remain the number one global killer in 2030. There are no established symptoms of the underlying disease of the blood vessels. However, heart attack or stroke may be the first sign of underlying disease. Heart attack or stroke is characterised by pain or discomfort in the centre of the chest, pain or discomfort in the arms, the left shoulder, elbow, jaw or back. (Boateng et al., 2017) (World Health Organization, n.d.).

Although CVD has a negative effect on a person quality of life and financial status, most of CVDs risk factors such as tobacco use, unhealthy diet and obesity, physical inactivity, harmful use of alcohol and air pollution are preventable (Boateng et al., 2017). Early detection is important for management with counselling and medicines (Zhang et al., 2017). However, gaps in knowledge of CVD conditions and their risk factors are important barriers in effective prevention and treatment (Zhang et al., 2017).

The primary problem this research seeks to tackle is the gap mobile apps plays in information access on CVD education and awareness, its symptoms and preventive measure. Mobile health (mHealth) is the use of mobile devices, such as mobile phones, portable monitoring devices, and personal digital assistants to support health practice. It involves using strategies such as smartphone apps, global positioning systems (GPS), and Bluetooth technologies. With approximately 500 million patients using mobile health (mHealth) applications to support their self-healthcare activities, information tailored on CVD education and its risk factors can effectively impact on reducing CVD complications (Cruz-Ramos et al., 2022).

The CardioRef project aims to bridge this gap. By developing a free android app, individuals or patients can access a general knowledge in CVD with information taken from trusted sources like the European Cardiology Society, America Cardiovascular Society and World health Organization. Utilizing animations in defining and describing CVDs, the project will provide an interesting and easy way to understand CVDs. With a weekly newsletter on CVDs, users of this app will be well informed on latest news and findings or updates on CVD management.

This research aim is not focus on diagnosis and treatment of CVDs or create conflicts with treatment regiments on CVDS but rather a platform to provide general knowledge on CVD and how to prevent or improve treatment through lifestyle changes.

1.2 Aims and Objectives

Aim:

The aim of this project is to increase education on CVDs, by providing a general information on the disease, risk factors, symptoms and preventive measures by using android mobile app to prevent and reduce complications and mortality of CVD.

Objective:

- **Background analysis**
 - Undertake a comprehensive review of existing literature on CVD and socioeconomic impact.
 - Evaluate previous studies and technologies used in developing app for CVD apps to determine methods to use.
- **Approach and design**
 - Build and design low fidelity prototyping UI of the sections of the app using Figma.
 - Implement scrum in the development which will help break down the user stories into tasks, monitor progress and testing at the end of each scrum
 - Create a robust backend infrastructure using Spring boot
 - Utilize MySQL to manage data.
- **Implementation**
 - Transform the design and prototype into a working application using Android Studio.
 - Use Java to implement the backend infrastructure and MySQL to store and retrieve data.
- **Testing**
 - Utilize the Android development function on an android phone to test each deliverable in a scrum.
 - Use tools in android studio to test app consistency, functional behaviour and usability.
- **Evaluation**

- Establish an evaluation criterion to measure the app effectiveness in providing easy and knowledge on CVD and its preventive measures.
- The criteria will also measure how easy users can use the app.

1.3 Project Approach

The project approach is designed an android app that provides general in information on CVD using both animation and text, a dictionary on some common terminologies and a weekly newsletter on CVDs. The methodology will ensure that the project's results are reliable, informative, and useful in providing insights into CVDs, preventive measure and how lifestyle changes can improve management of CVDs.

1.3 Dissertation Outline

Initially, the project will undertake a comprehensive literature review to identify existing knowledge and research gaps in CVDs education and awareness and the role of mobile apps. This will help to formulate the research questions and objectives, guiding the subsequent phases of the project.

After the research, the project will focus on developing an android-base app. This app will serve as an educative tool for providing trusted knowledge and information on CVDs, its preventive measures and lifestyle changes to improve the disease management. The more easily the users can understand the disease using animated video and dictionary, the more beneficial the app will be.

To ensure relevance of the information, the app will have its source of information from trusted websites like World Health Organization, European Society of Cardiology and America Cardiovascular Society. A user-friendly interface will be designed to facilitate easy access and navigation through the app various sections. These sections include an animated videos on CVDs, a dictionary for common CVD terminologies and a newsletter. This phase aims to provide an understanding of CVD, its preventive measures and a general lifestyle to improve CVD management.

An iterative methodology will be used for the duration of the project, enabling ongoing improvement of the app, based on user feedback. This will ensure that the project remains adaptive and responsive to emerging insights and challenges.

In summary, the project will be carried out using a methodological strategy that incorporates theoretical research, technical development, and trusted information source.

This comprehensive methodology will enable the drawing of reliable conclusions on how effective mobile apps can help in the prevention and management of CVDs.

2 Background

2.1 Overview of impact of Noncommunicable (chronic) disease knowledge and awareness on mortality reduction

According to WHO, noncommunicable diseases (NCDs), often referred to as chronic diseases, are characterized by their prolonged duration and arise from an interplay of genetic, physiological, environmental, and behavioral influences. Non-communicable diseases (NCDs) primarily comprise of cardiovascular conditions, including heart attacks and strokes, various forms of cancer, chronic respiratory illnesses such as chronic obstructive pulmonary disease and asthma, as well as diabetes. The disease has a significantly greater impact on individuals residing in low- and middle-income nations, accounting for over 75% of the total global deaths attributed to NCDs, which amounts to 31.4 million fatalities (World Health Organization, 2024).

In 2014, it was disclosed that school children in rural regions of India had a significantly limited understanding of the risk factors linked to non-communicable diseases (NCDs) and a lack of awareness regarding preventive measures for these conditions (Anju Ade et al., 2014). This considerable deficiency in knowledge about NCDs and their associated risks contributes to unhealthy lifestyle choices, such as physical inactivity, excessive television viewing, insufficient consumption of fruits and vegetables, and high intake of junk food. Consequently, this behavior has resulted in a rise in overweight and severe obesity, which are recognized risk factors for non-communicable diseases (Rahamathulla and M, 2020).

These findings provide a concise understanding of how awareness and knowledge on CVDs can help WHO 2030 agenda for sustainable development as it recognizes NCDs as a major challenge for the program (World Health Organization, 2024). This underscores the significance of pertinent research in this domain and thereby reinforcing the rationale for this project.

Additionally, it is crucial to consider the issue from the viewpoint of people who are at high risk, as this perspective is essential for developing the most effective solutions. Mobile app has been selected as a focus area due to the widespread prevalence of mobile devices as the primary means of communication among individuals today. Consequently, it is posited that mobile health (mHealth) represents a valuable instrument for encouraging healthy lifestyles and behaviors.

2.2 Definition and types of CVD

Numerous definitions and characterizations of CVD in contemporary literature. The variations in these definitions often arise from the specific context or intent behind their usage.

Additionally, the complexity of CVD definition is compounded by the fact that individuals experience varying degrees of impact from the condition. Below are several definitions sourced from diverse references. NHS UK (NHS, 2022) defines CVD as “a general term for conditions affecting the heart or blood vessels. It’s usually associated with a build-up of fatty deposits inside the arteries (atherosclerosis) and an increased risk of blood clots. It can also be associated with damage to arteries in organs such as the brain, heart, kidneys and eyes.” NHS UK further stated that CVD represents a significant contributor to mortality and morbidity in the United Kingdom; however, its incidence can frequently be mitigated through the adoption of a healthy lifestyle.

The British Heart Foundation (British Heart Foundation, 2019) defines CVD as “an umbrella name for conditions that affect your heart or circulation. These include high blood pressure, stroke and vascular dementia.” The definition from Britannica Encyclopedia (Encyclopedia Britannica, n.d.) is “any of the diseases, whether congenital or acquired, of the heart and blood vessels. Among the most important are atherosclerosis, rheumatic heart disease, and vascular inflammation. Cardiovascular diseases are a major cause of health problems and death.” History and science behind CVD can be obtain from Britannia website.

Based on the conducted research, I have developed the following understanding of CVDs: CVDs are related to blood vessels. These blood vessels take deoxygenated blood from organs to the heart and carries oxygenated blood from the heart to the organs including the heart itself. When these blood vessels passage gets thinner due to accumulated low lipid lipoproteins on its walls known in medical terms as atherosclerosis, it restricts the supply of oxygenated blood to the organs and the heart itself. This can lead to complications such as high blood pressure, heart attack, stroke and damage to organs. These conditions when not medically attended to can lead to death. Other causes of CVD anomalies are congenital conditions that are present at the time of birth, dysfunctional heart valves and arrhythmia (irregular heartbeat problems) (Cleveland Clinic, 2022).

The underlying vascular disease may present without any noticeable symptoms. In some cases, a heart attack or stroke could serve as the initial indication of such a condition. Common symptoms associated with a heart attack includes pain or discomfort located in the central region of the chest, as well as pain or discomfort that may radiate to the arms, left shoulder, elbows, jaw, or back (World Health Organization, n.d.). Individuals may also encounter challenges such as breathing difficulties or shortness of breath, along with symptoms like nausea or vomiting, feelings of light-headedness or faintness, cold sweats, and pallor. It is

noteworthy that women are more prone than men to experience shortness of breath, nausea, vomiting, as well as discomfort in the back or jaw (World Health Organization, n.d.).

Cardiovascular diseases comprise of a wide array of conditions, among which are the following: (Cleveland Clinic, 2022)

- Arrhythmia refers to disorder affecting the heart's electrical conduction system, resulting in irregular heart rhythms or rates.
- Valve disease refers to the dysfunction of heart valves, which are critical structures that facilitate the flow of blood between the heart's chambers and into blood vessels.
- Coronary artery disease involves complications within the heart's blood vessels, often characterized by obstructions.
- Heart failure denotes an impairment in the heart's ability to pump or relax effectively, resulting in fluid accumulation and difficulty in breathing.
- Peripheral artery disease pertains to abnormalities in the blood vessels supplying the arms, legs, or abdominal organs, typically manifesting as narrowing or blockages.
- Aortic disease are issues related to the aorta, the major vessel responsible for transporting blood from the heart to the brain and the rest of the body, including conditions such as dilation or aneurysms.
- Congenital heart disease refers to structural heart defects present at birth, which can impact various components of the heart.
- Pericardial disease involves disorders affecting the pericardium, the protective lining surrounding the heart, including conditions like pericarditis and pericardial effusion.
- Cerebrovascular disease is characterized by problems in the blood vessels supplying the brain, often resulting in narrowing or blockages.
- Deep vein thrombosis (DVT) is a condition marked by the obstruction of veins that return blood from the body and brain to the heart.

Cardiovascular diseases are largely preventable through the modification of behavioral and environmental risk factors, including the cessation of tobacco use, the adoption of a healthy diet, the management of obesity, the promotion of physical activity, the reduction of harmful alcohol consumption, and the mitigation of air pollution (World Health Organization, n.d.).

2.3 Research on CVD knowledge and awareness

Cardiovascular disease is the leading cause of death worldwide and represents a major obstacle to the sustainable development goals of the WHO 2030 Agenda. Therefore, awareness and education about the disease is crucial. In the past three decades, developed or high-income

countries have recorded a major decline in CVDs. Adoption of population-wide preventive strategies, efficient primary and secondary preventive healthcare, and the availability of better acute event treatment has been a backbone for this achievement (O'Flaherty, Buchan and Capewell, 2012). Despite this development, developing countries or low- and middle-income countries have recorded the opposite (Boateng et al., 2017). Below are some research papers relating the rise of CVDs to knowledge and awareness.

A review of data and articles conducted in accordance with the 2017 PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines aimed to assess the knowledge and awareness of cardiovascular diseases (CVDs) in Sub-Saharan Africa (SSA). The findings indicated that a significant barrier to health-seeking behaviors in the region is the persistent lack of understanding about CVDs and their associated risk factors. Additionally, the study highlighted that formal workers demonstrated a higher level of awareness and knowledge about CVDs and their risk factors compared to individuals from rural and urban households (Daniel et al., 2017).

Furthermore, research conducted by Baig et al. (2015) examined the rates of obesity and hypertension among university students in Jeddah, Saudi Arabia, while also evaluating their understanding and perceptions of cardiovascular disease (CVD) risk factors. The findings reveal a significant disparity between students' knowledge of CVD risk factors and their actual health-related behaviors. The authors stress the importance of implementing thorough awareness initiatives to close this gap and encourage healthier lifestyle choices among young adults.

Research studies such as "Awareness of cardiovascular disease among Korean women: Results from a nationwide survey" (Kim et al., 2022) and "Impact of a simple educational intervention on awareness regarding cardiovascular disease among school-going adolescents in a rural area of Bengaluru district, India" (Coates et al., 2023) presented results that were consistent with the findings of the two previous articles.

In summary, increasing awareness and providing education regarding cardiovascular disease (CVD) and its associated risk factors is crucial in the battle against this condition. Given the widespread use of smartphones, leveraging mobile applications to disseminate this information can significantly contribute to reducing mortality rates and the prevalence of the disease.

2.4 Research on mobile apps on CVD awareness

Mobile applications have become an important area of research delving into their efficacy for raising the awareness and encouraging preventive behavior against cardiovascular diseases (CVDs). There is growing appreciation that such applications have an important role to play in the promotion of awareness, education and in motivating the individuals to take preventive actions. [\(Fabian et al., 2024\)](#) [\(Erin et al., 2021\)](#). CVD is becoming increasingly prevalent around the world, hence the necessity for new and accessible intervention potentials, and mHealth is an avenue that shows promise for reaching huge populations [\(Islam and Maddison, 2021\)](#). The WHO has high-lighted a global burden for the diseases related to CVD specifically demanding some effective leadership in their prevention strategies [\(Who.int, 2024\)](#).

Mobile health interventions are said to help increase the awareness of CVD and change people's behavior towards it, but the effectiveness of these interventions varied widely depending on several factors, including the design of the app, prescription and engagement methods, characteristics of the target population, and integration with existent health systems.

Several studies have shown that CVD awareness apps are effective in improving knowledge and attitudes towards CVD risk factors and prevention [\(Coorey et al., 2018\)](#). Thus, such improved knowledge can become a good base for better behavioral changes.

Increased Awareness: The use of apps has increased awareness of CVD risk factors and prevention. Greater awareness through app use could very much move users to healthier lifestyles. Merely raising awareness might not influence lasting behavior changes.

In a related study featured in the JMIR Cardio journal, it was shown that users of applications who had both educational materials and interactive components, like pictures, quizzes, and videos, performed much better regarding CVD risk factors. These users who used these sections understood more about how their practices of engaging in certain activities affected their heart [\(Erin et al., 2021\)](#).

Improved Knowledge: Apps can deliver to the public evidence-based education very well and improve understanding by the users about CVD risk factors, prevention, as well as the lot importance of adherence to medications [\(Xie et al., 2017\)](#). This new knowledge can enable users to make informed decisions regarding their health. In fact, engagement strategies are also very important for successful educational content delivery.

Change in Attitudes: Some signs show that using these applications can help improve the attitudes of people toward healthy choices and increase their self-efficacy for adopting healthy behaviors. Such attitude changes are good boons toward behavior change. Measuring even gradual changes in attitudes, however, is not an easy task and must be done with specific instruments.

The varying effect of CVD awareness apps on the changes in behavior is dependent upon targeted behavior and the specific app design. **Engaging Everyone in Physical Activity:** App users demonstrated improvements in physical activity levels (Bushey et al., 2024). The effect size differs according to app design, engagement strategies, and intervention intensity. Objective measures of physical activity, such as accelerometry data, are warranted for verification of self-reported changes.

Diet: Evidence from Some studies shows positive changes in behavior relating to fruit and vegetable intake (Krzyzanowski et al., 2020). This is why there is no conclusive evidence of effect. More effective strategies are therefore needed to bring about maintained changes in diet. There exist objective methods whereby food diaries or 24-hour recalls could be done: Self-report changes need confirmation by Objective dietary assessment methods.

Medication Adherence: CVD Medication therapy apps have been rewarding. They remind patients to take their medication and educate them on the importance of compliance. They, however, depend on the design of the reminder system and user engagement on their effectiveness.

Engagement of user and adherence to it have shown to be the most critical factors when concerning the effectiveness of CVD awareness apps. There are a lot of obstacles which can inhibit the use of the app and bring about continued change in behavior. These technologies are limited in low-resource environments by the absence of cellphones or even reliable internet access. Moreover, using internet platforms might be challenging for older persons, who are more susceptible to cardiovascular ailments. Many apps which are available in the market are not scientifically validated, and this decreased their political value as well as the importance. A systematic review showed that 80% of the apps assessed, had not been tested in peer review or supported by any medical authorities.(Fabian et al., 2024). In the absence of professional approval, the users may become victims of misinformation which can in turn compel the user to make decisions that are detrimental to their health.

Mobile applications have a lot of potentials when it comes to raising awareness of CVD, encouraging healthier habits, and eliminating health disparities. To realize those potentials, tackling some key challenges is in order, including app design, user engagement, integration with

the healthcare system, and health equity. Future research needs to consider longitudinal studies, proof-of-concept trials, defining qualitative insights, clinical integration, health equity-aspects, and standardized evaluation tools-all in the quest for making the most of mHealth interventions in CVD prevention. By overcoming these diverse challenges, mHealth has the potential to transform the landscape of cardiovascular health in the world and to mitigate the resounding burden of CVD. Such apps may be culturally relevant, more user-friendly, and integrated into clinical practice, holding great promise for public health and health equity. There are great expectations soon regarding prospects offered by technological advancement in AI and its application in healthcare.

2.5 CVD awareness-based mobile apps

The substantial global health burden attributed to cardiovascular diseases (CVD) highlights the necessity of preventative measures to lessen their effects. Because they provide creative platforms to spread knowledge, encourage healthy habits, and raise awareness about CVD, mobile applications have emerged as essential instruments for public health campaigns. By providing individualized material on risk factors, lifestyle modifications, and disease management techniques, these applications serve a wide range of users, including caregivers, healthcare professionals, and those at risk [\(Fabian et al., 2024\)](#) [\(Erin et al., 2021\)](#).

Most successful apps share similar characteristics, which happen to be risk assessment tools [\(Weichelt et al., 2021\)](#), [\(Buss et al., 2021\)](#), [\(Ijsirt.com, 2023\)](#). This feature allows users to quantify their personal CVD risk according to diverse parameters: age, sex, family history, lifestyle choices, and clinical measure. Such tools provide much of the initial interaction to users as impetus to make further inquiries about their risk levels and encourage them to undertake measures that improve cardiovascular health. Educational modules are other common such features, [\(Buss et al., 2021\)](#), [\(Ijsirt.com, 2023\)](#), educating the users on evidence-based CVD risk factors, prevention, and healthy lifestyle choices. These can be anything from simple text-based information to engaging videos and quizzes tailored to various styles of learning.

Personalized goal setting is one among many features in CVD awareness apps [\(Buss et al., 2021\)](#), [\(Burke et al., 2015\)](#), allowing users to determine specific, measurable, achievable, relevant, and time-bound goals (SMART) related to their own contexts and desires. This personalization makes the user more motivated and feel more responsible and in control of the process that will help improve cardiovascular health. Behavioral patterns tracking is another key aspect [\(Coorey et al., 2018\)](#), [\(Buss et al., 2021\)](#), [\(Krzyzanowski et al., 2020\)](#) for measuring the progression towards goal achievement by users. Tracking may involve tracking levels of physical activity, food intake,

adherence to drugs, or other health-related behaviors. Viewing progress made over time is an important motivational factor that encourages users to continue engaging with the app.

Gamification strategies (Weichelt et al., 2021) are increasingly employed to enhance user motivation and engagement. These strategies involve awarding points, badges, or other incentives for achieving specific goals, which fosters a greater sense of accomplishment and encourages ongoing participation. To further enhance engagement and motivation, it is beneficial to incorporate social features that allow users to connect with one another, share their progress, and offer mutual support (Clifford et al., 2021). Additionally, multilingualism (Ijsirt.com, 2023) plays a crucial role in promoting accessibility and inclusivity, enabling a broader audience to utilize the application. The integration of data from wearable sensors (Kozik et al., 2021) (Burke et al., 2015) (Eapen et al., 2016) (Huang et al., 2022) is expected to provide a more objective and comprehensive means of monitoring health behaviors and associated physiological metrics. However, this also raises concerns regarding data security, privacy, and the potential for false or inaccurate information.

Effectiveness and Impact on CVD Risk Factors examines the evidence regarding the role of cardiovascular disease awareness applications in altering risk factors associated with cardiovascular disease. Numerous studies investigate how these applications influence weight management (Buss et al., 2021) (Burke et al., 2015). The findings vary, with some studies indicating success while others report no significant changes. Certain research indicates a notable difference in weight and body mass indices (BMIs) between users of CVD awareness apps and non-users, although not all studies corroborate this observation. These discrepancies often stem from variations in app design, the intensity of interventions, participant demographics, and research methodologies. Similarly, studies analyzing the impact of these applications on physical activity (Spaulding et al., 2021) (Kozik et al., 2021) (Krzyzanowski et al., 2020) differentiate between participants who engage with the app and those who do not. Some research highlights a significant increase in physical activity among app users, while others report minimal or no effects.

The influence of cardiovascular disease (CVD) awareness on nutritional behavior, as noted by Spaulding et al., 2021 and Krzyzanowski et al., 2020, continues to be a topic of research. Some scholars argue that these applications serve as effective tools for promoting healthier eating habits, while others contest their efficacy. It is evident that the design of the application's dietary features, the degree of user engagement, and the methods of intervention can significantly impact dietary behavior outcomes. Additionally, a critical area of focus in CVD prevention is smoking

cessation, with various studies exploring the effects of mobile health (mHealth) interventions in this context (Kozik et al., 2021) (Ghavami et al., 2024). Some research indicates a substantial and statistically significant increase in smoking cessation rates among app users, while other studies report minimal effects, and some find no impact whatsoever. The success of these interventions is also influenced by the specific behavior change techniques (BCTs) employed, the level of user participation, and the broader context in which the intervention occurs.

User adherence to an application refers to the consistency with which individuals utilize the app over a period (Krzyzanowski et al., 2020). Generally, improved health outcomes are associated with higher levels of adherence. Several factors influence user engagement with the application, including its design, functionality, personalization, social support, and accessibility (Clifford et al., 2021) (Spaulding et al., 2021). Applications that feature intuitive designs, user-friendly interfaces, and tailored options tend to promote ongoing usage. Users are more likely to engage with and remain committed to the app if it includes social elements, such as the ability to connect with others and share personal experiences. Additionally, the effectiveness of these applications is enhanced when they are designed to accommodate a wide range of users, each with varying technological skills and levels of health literacy.

Mobile applications designed to enhance awareness of cardiovascular disease (CVD) offer a potential solution to global health challenges by empowering individuals to manage their heart health proactively. These applications equip users with the necessary tools to make informed choices through a combination of risk evaluations, behavioral change strategies, and educational resources.

3 Methodology

This chapter discusses the method that would be used to achieve the project's objectives. This covers a discussion on the software methodology, its framework and the rationale behind this decision.

3.1 Scrum Software Methodology

The CardioRef project builds a reference android mobile application to raise awareness and educate people about CVDs using the scrum development methodology.

Scrum is an agile software development methodology that manages product development in an incremental, iterative and timely manner. One of the fundamental tenets of scrum is requirements volatility, which acknowledges that customers may alter their minds about what they need and want during the production process. These unforeseen difficulties are difficult to handle in a conventionally planned or forecast way, which is why scrum/agile methodology has an edge.

Scrum utilises an empirical approach, acknowledging that the problem cannot be fully described. Instead, it concentrates on adapting to changing market conditions, expanding technologies, and emerging requirements (Sakshi Sachdeva, 2016). Additionally, with features like customizable requirement selection for sprints and no set procedures to follow, Scrum provides a tailored approach to working on various projects with a range of requirements (Srivastava, Bhardwaj and Saraswat, 2017). Scrum makes sure that time and resources are used efficiently. Big projects are broken up into manageable sprints. There is a sprint review at the end of each sprint where developed codes are tested. Scrum focusses more on the team's interactions with the organisation and with each other. However, with my experience in project management, scrum framework would be appropriate for this project taking the size of the project and the time of completion into consideration.

3.2 Scrum framework and artifacts

The scrum framework prioritizes value and high visibility of progress when guiding the creation of a product. Three scrum roles make up the framework:

- **Product owner:** Product owner is responsible for optimizing the value of the final product that the Scrum Team produces.
- **Scrum master:** The Scrum Master functions as a Servant Leader within the Scrum team framework. This individual directs the team alongside the Product Owner, ensuring that all members adhere to the prescribed Agile methodologies with fidelity.

- Development team: Development team individuals within the Scrum Team who are dedicated to producing any component of a functional Increment during each Sprint
- However, since I would be the only one developing this project, I will be performing all these three roles.

Agile scrum artifacts are “information that a scrum team and stakeholders use to detail the product being developed, actions to produce it, and the actions performed during the project. The main agile scrum artifacts are product backlog, sprint backlog, and increments” (Harris, n.d.). The main artifacts of scrum are:

- Product Backlog: it is the ordered list of all the user stories needed to be done to complete the mobile app. It is derived from the app requirement.
- Sprint Planning: this is where user stories for a sprint is selected.
- Sprint Backlog: this contains the user stories for the sprint. These are the user stories that will be selected in the sprint planning.
- Sprint: is the set period of where sprint backlogs will be completed. It is time-boxed and would last for two weeks.
- Increment: this is where previously built and newly built deliverables are all integrated, tested and ready for deployment
- Sprint Review: this is where the team present their work to stakeholders and get feedback. A demo session would be help at the end of each sprint with my personal tutor for feedback.
- Sprint Retrospective: based in the feedback from my personal tutor, I would to personal evolution as part of plans to improve the next sprint.

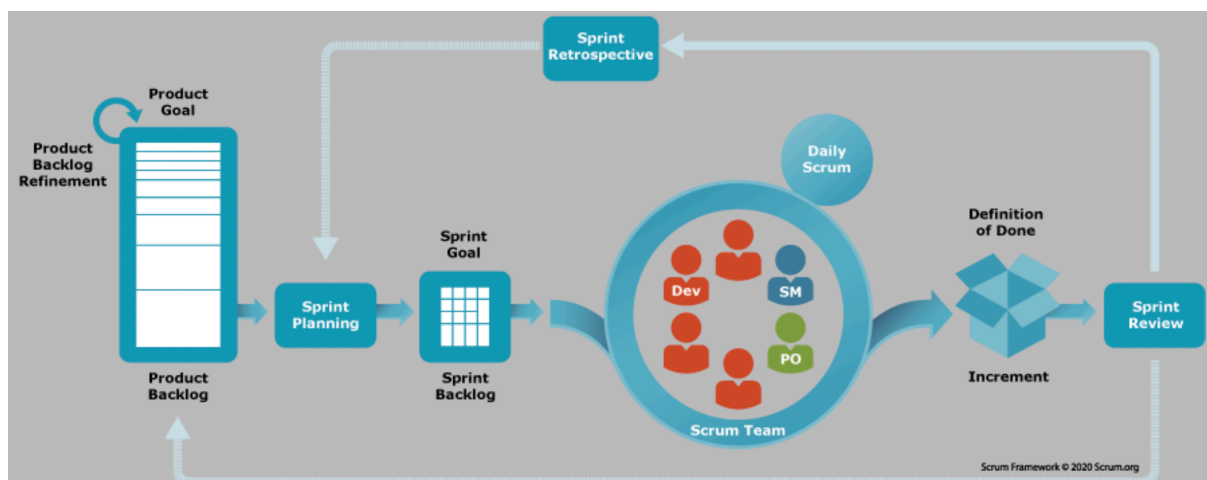


Figure 1 An image show scrum artifact.

4 Design

This section discusses how CardioRef application is designed. This section covers project requirements, user stories, the design process, and prototyping which is needed for the methodology discussed in chapter 3.

4.1 User stories

In Scrum, user stories are a summary of a software feature or functionality written informally and from the viewpoint of the end user. It describes the type of user, what they want from the system and why. The user stories for the application are as follows:

- As a user, I want the user interface to be easy to understand so that I would be able to use the app easily
- As a user I want to access a video animation of the types of CVDs so that it can help me understand the disease well
- As a user I want to access the text-based information on the type of CVD so that I can read more about the disease
- As a user I want to access dictionary of common terminologies of CVD so that I it can help me clarify or understand CVDS
- As a user I want to access the weekly newsletter on CVD so that I can stay up to date on CVD

Use case diagram was developed based on the user stories to better illustrate the user and tasks involved.

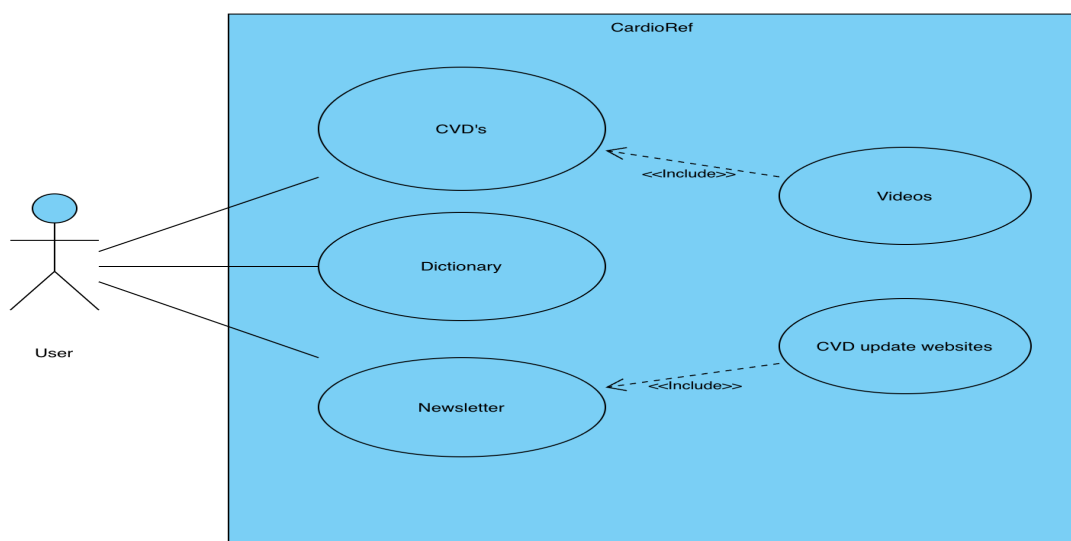


Figure 2 Use case diagram for CardioRef

A sequence diagram for the CVDs feature on the application is shown below.

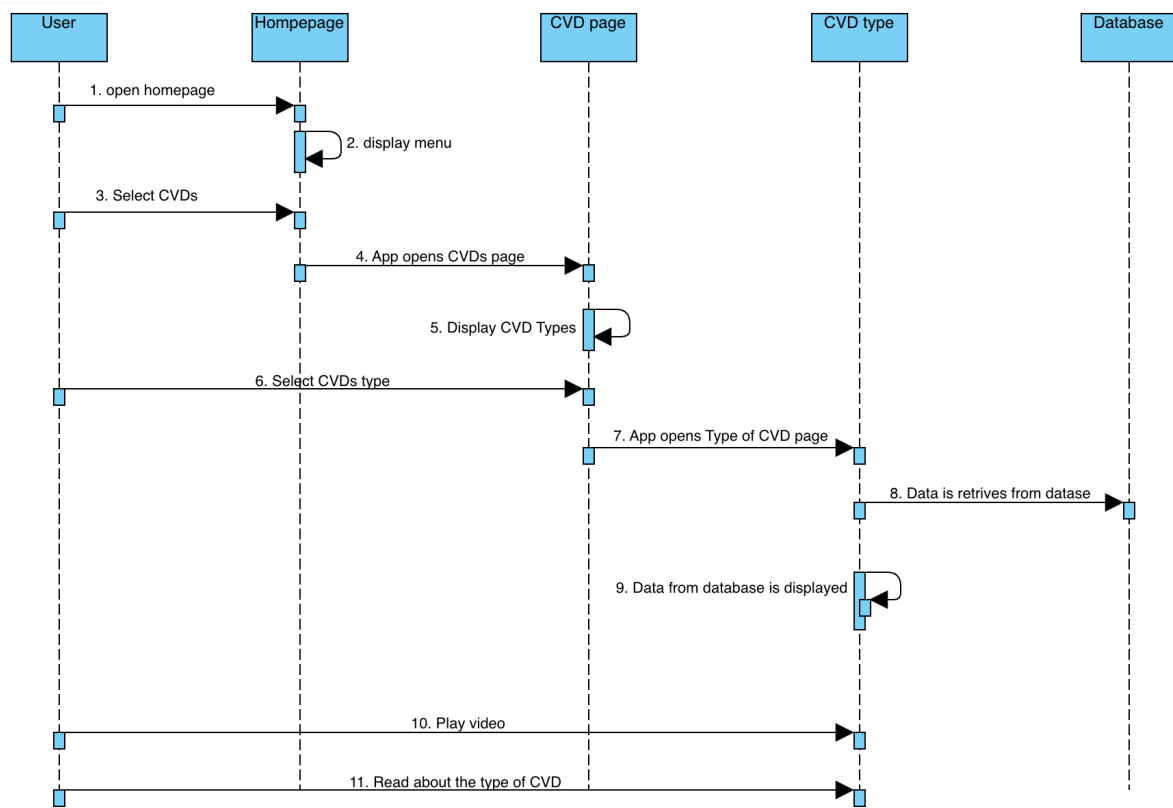


Figure 3 Sequence diagram for CVD feature of the app

4.2 Project Requirements

The project requirements were derived from the project aim and objectives. The application is intended to provide a general knowledge and education on CVDs. Below are the functional and non-functional requirements of the application:

Functional Requirements

This are statements the describe what a system should do or what should be implement for users to achieve their goal. Functional requirements of CardioRef are as follows:

- The application should be able to play videos on CVDs
- The application should be able to display textual information on CVDs
- The application should be able to give up to date information in CVDs
- The application should be able to give meaning of medical terminologies relating to CVDs

Non-functional Requirements

- The application should be able to hold both high resolution video and text

- The application should be able to import newsletters from trusted websites
- The application should be able to do a search on both newsletter page and dictionary page

4.3 Prototyping

To examine and test the functional requirements found during the requirement analysis process, a high-fidelity prototype of CardioRef was created using Figma.

In addition of being the first model, prototype is used to test and refined the design and construction of a product before it is into production. The screenshots of the prototype created with Figma are displayed in the picture below. This prototype was tested using 412 x 917 Android compact screen.

Figure 3 shows the Home Page User Interface. CardioRef logo and three option buttons are featured on the Home Page. The user is directed to the CVD's page by the first button, the Dictionary page by the second button and the Newsletter page by the last button.



Figure 4 CardioRef Home UI Figma Prototype

Selecting the CVD button directs you to the CVD page, where various types of CVDs are displayed. This page provides a list of CVD types along with the topics that organize related information.

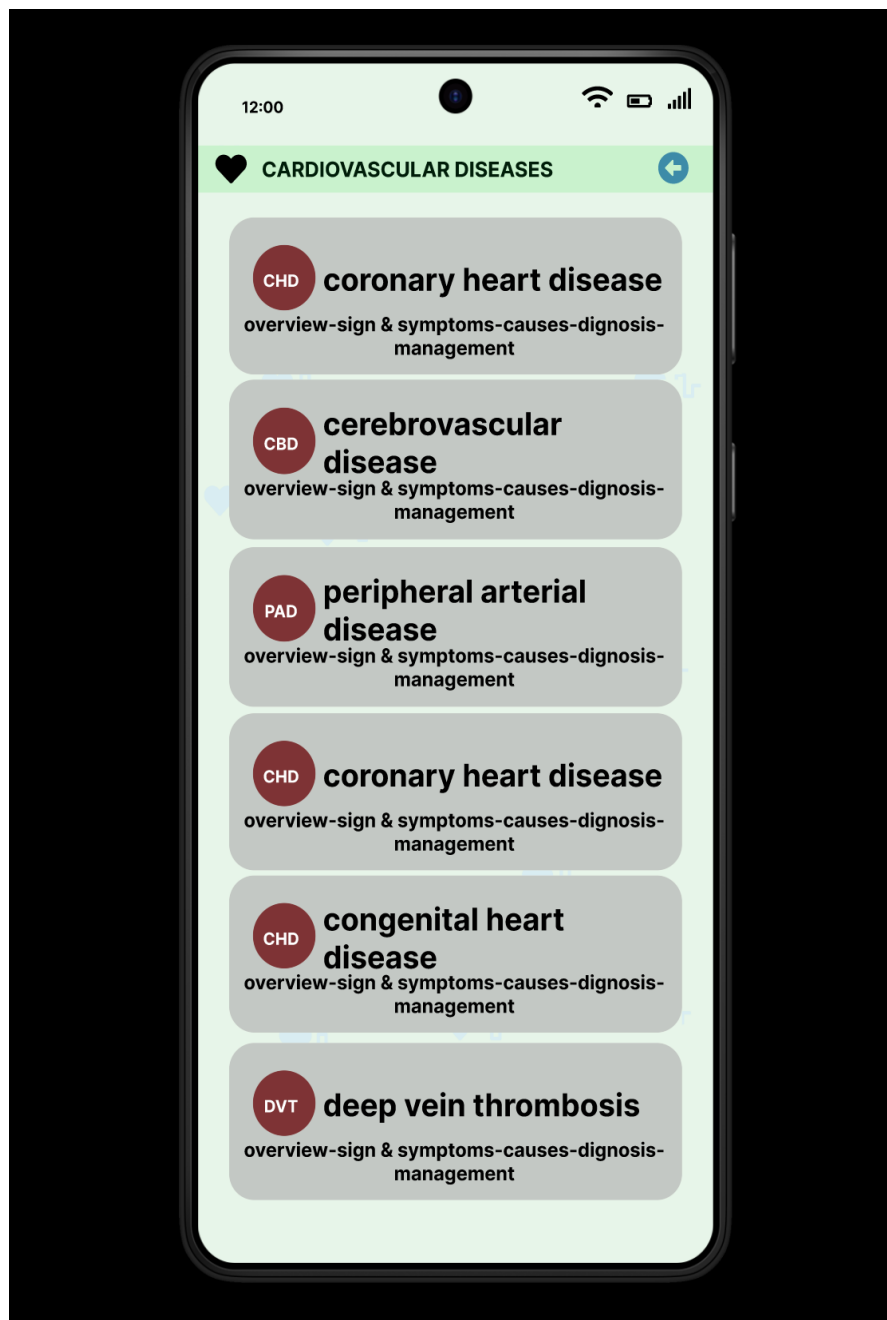


Figure 5 CardioRef CVD UI Figma Prototype

Every category of CVD is represented as a clickable button that leads to a comprehensive page. These pages include an animated video related to the chosen CVD and provide a summary of the subtopics highlighted on the primary CVD page.



Figure 6 CardioRef Selected CVD UI Figma page Prototype

The second button located on the homepage is the Dictionary button, which directs users to the Dictionary page upon selection. This page features a compilation of frequently used medical terminology associated with cardiovascular diseases (CVDs). When a user clicks on any term, it expands to reveal its definition. Additionally, a search button is provided on this page to enhance the ease of locating specific words.

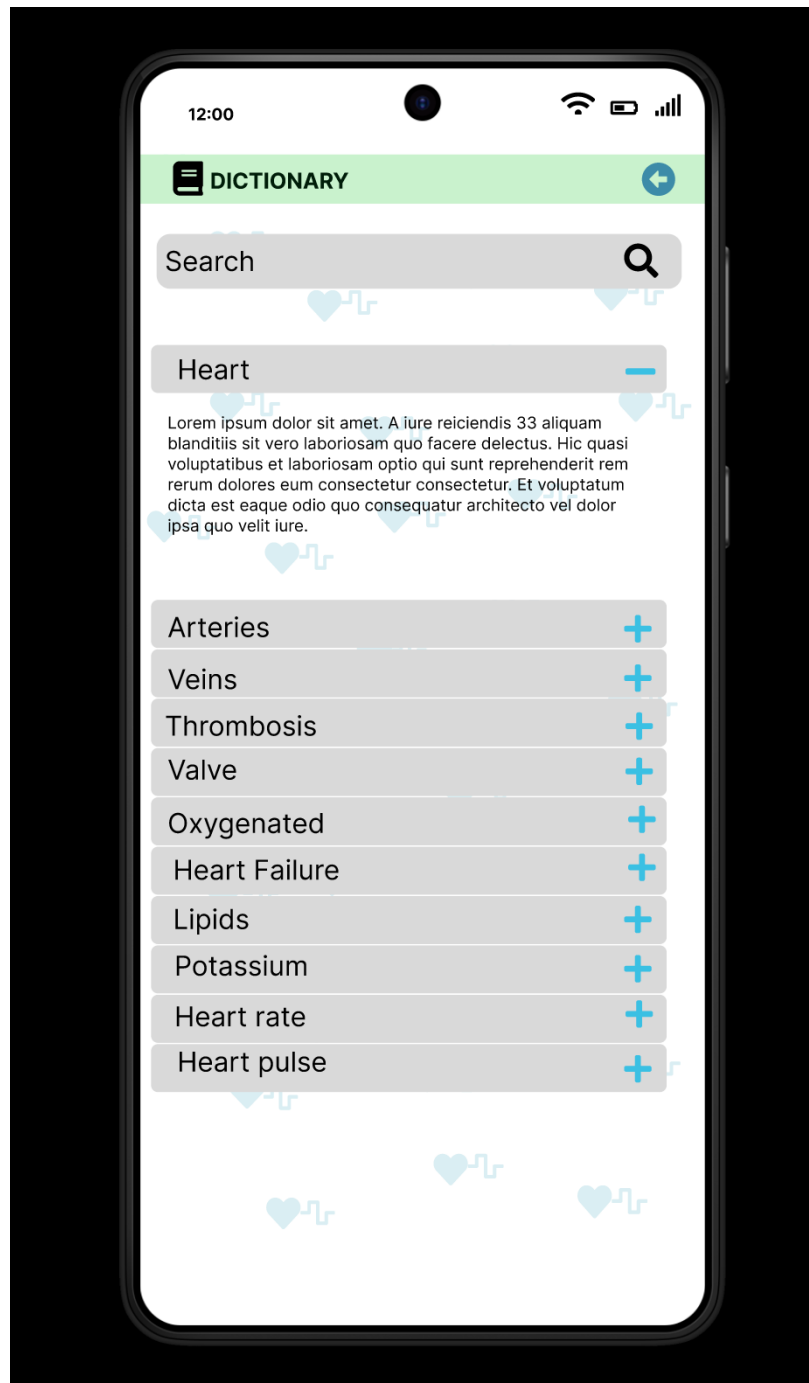


Figure 7 CardioRef Dictionary UI Figma Prototype

The third button located on the homepage is designated for the Newsletter. This page features articles and updates concerning cardiovascular diseases (CVDs). It includes the source of each news item or article, along with the date it was published or last updated. By clicking on the headline, users are redirected to the original website of the article or news in their device's default browser.

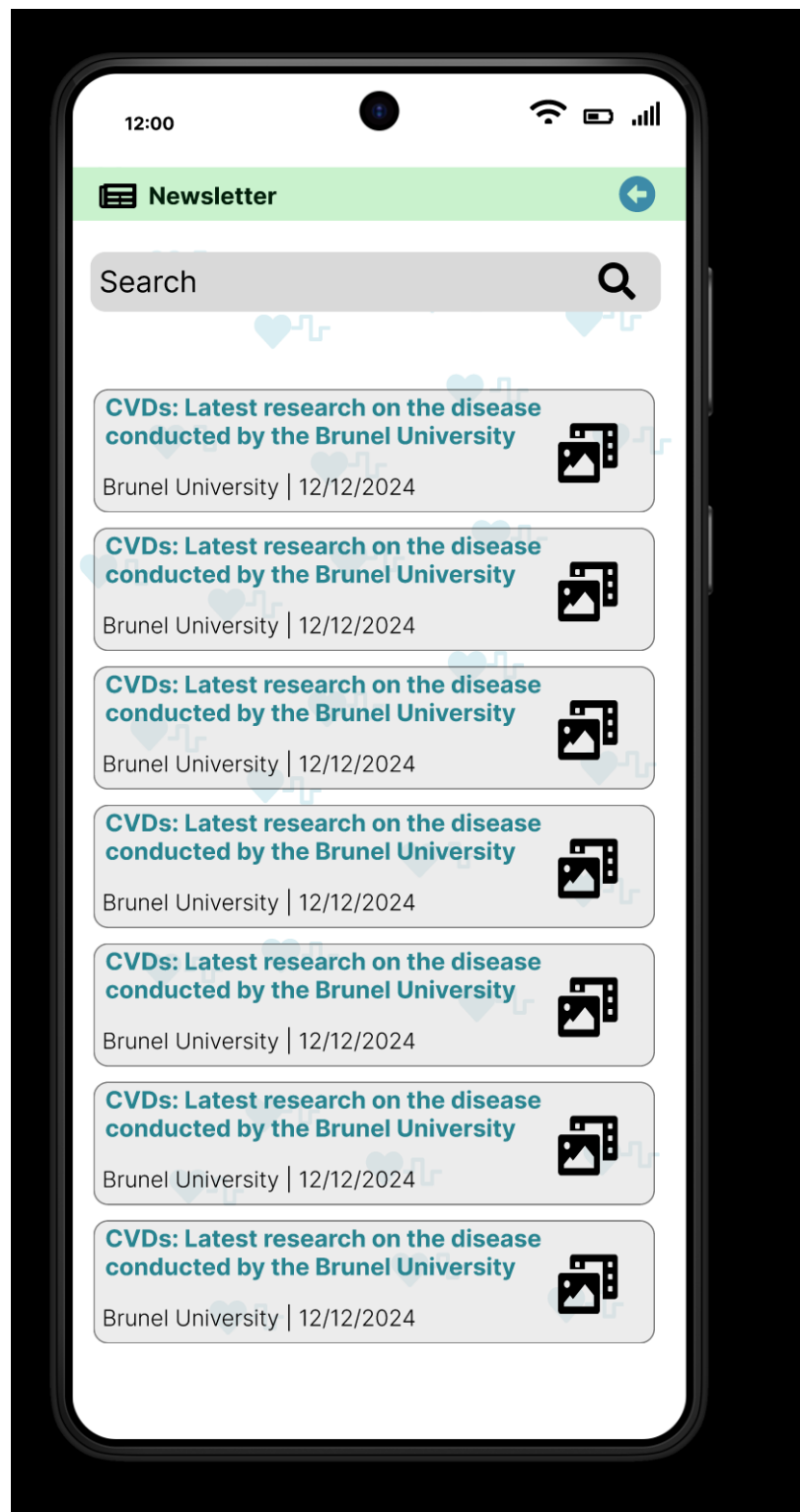


Figure 8 CardioRef Newsletter UI Figma Prototype

Link to CardioRef App Figma Prototype will be found at appendix II

5 Implementation

This section will analyze the project's implementation/execution. It will outline the methodology from chapter three, the technologies and tools utilized, and the action taken to create the proposed mobile application. Additionally, it will address the risk management during this stage.

5.1 Technologies and tools

The software architecture comprises of a Mobile Application Layer, System Management Layer and Data Layer.

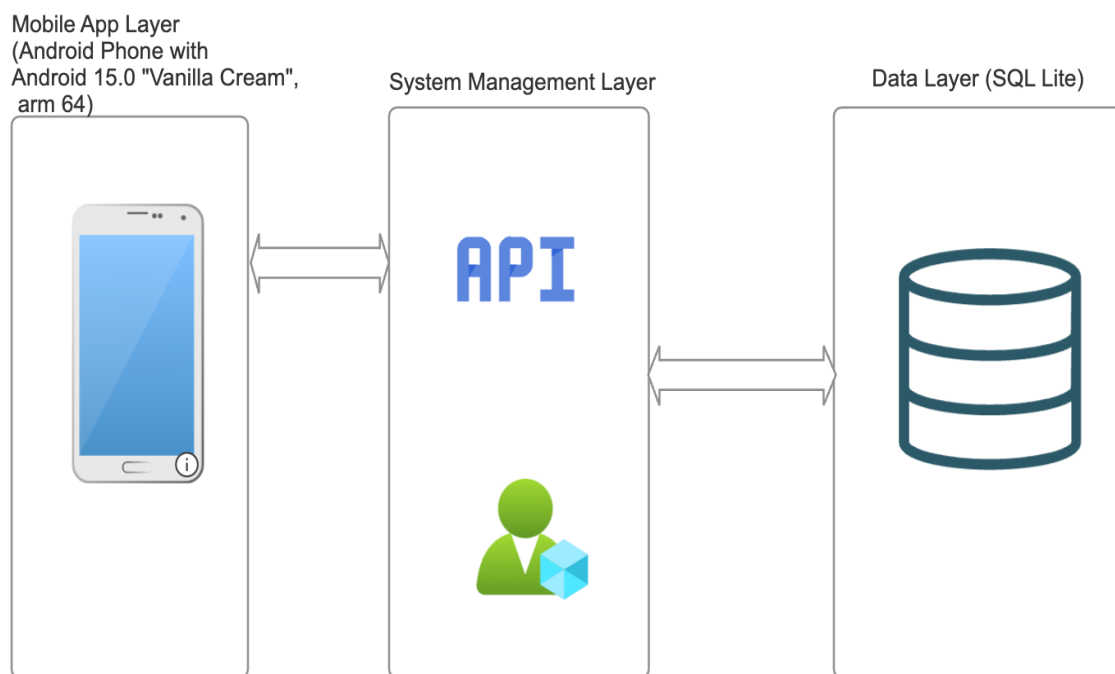


Figure 9 The architecture of the application showing the various layers

The Application Layer was created utilizing Android Studio Ladybug | 2024.2.1 Patch 3, featuring Runtime version 21.0.3+-79915917-b509.11 aarch64 and the OpenJDK 64-Bit Server VM developed by JetBrains s.r.o. During the development phase, testing was conducted by executing the application on a Pixel 9 API emulator running Android 15.0 ("VanillaIceCream") and arm64 architecture.

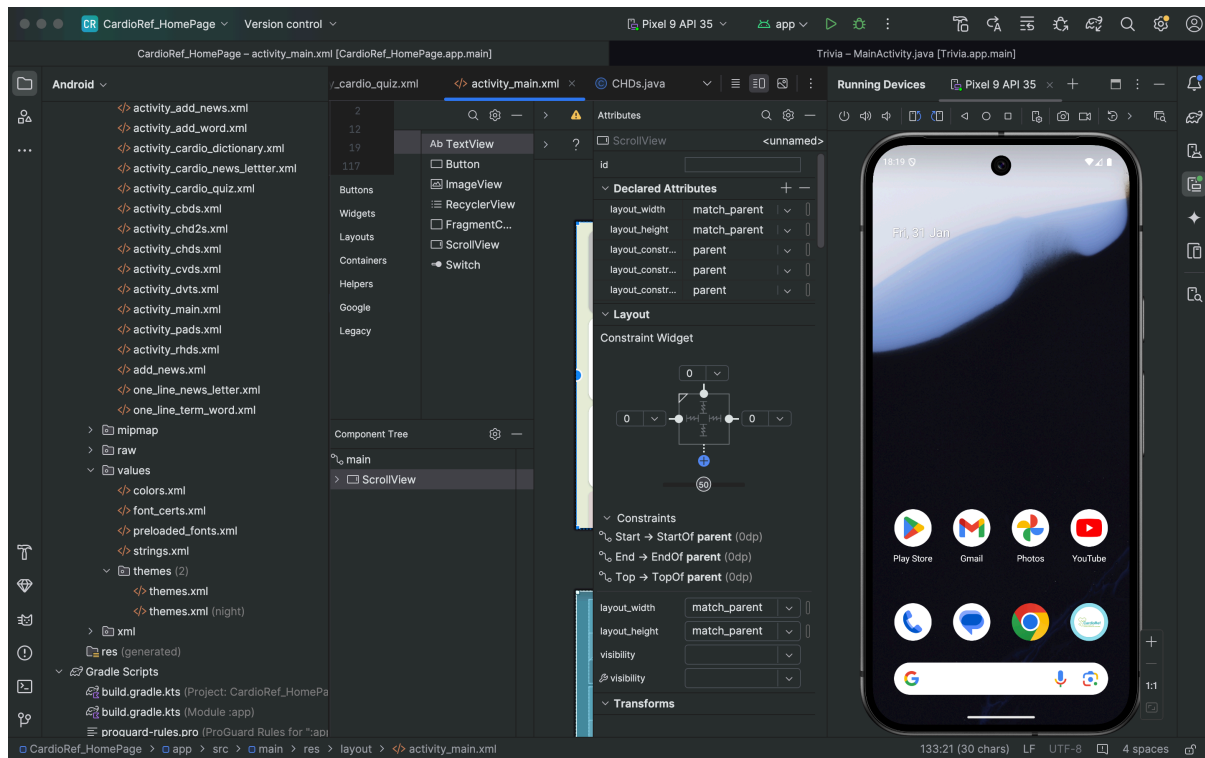


Figure 10 A screenshot of Android Studio Ladybug with Pixel 9 emulator

The System Management Layer includes an API designed for the creation of database tables. The necessary APIs for utilizing the database within Android Studio can be found in the "android.database.sqlite" package. This layer encompasses CRUD functions that facilitate data management, although these functions are not visible to the user, as the application provides only a user interface.

The Data Layer is comprised of SQLite, a free relational database that enables users to perform database operations on Android devices. This functionality includes the storage, management, and retrieval of long-term data. SQLite is integrated into Android by default, which means there is no requirement for additional database setup or administrative tasks (Shivani Patel, 2023). The SQL Lite App was utilized to oversee the structure of the created tables.

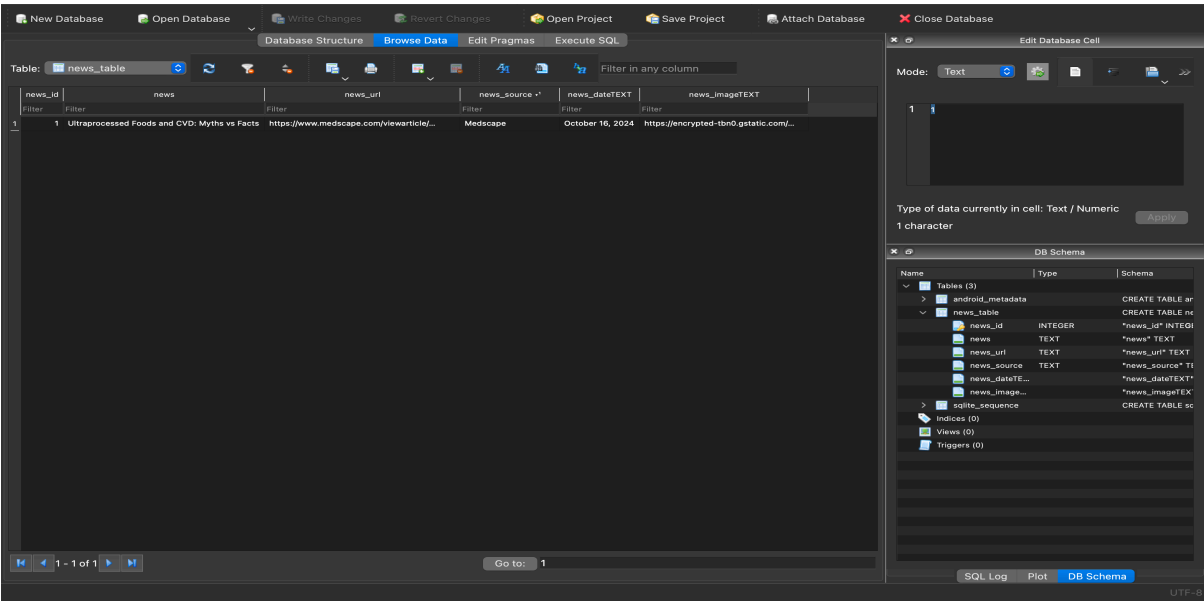


Figure 11 A screenshot of SQL Lite database application

5.2 App Development Methodology

As outlined in chapter three, the development of the entire application employed the Scrum methodology. This approach is characterized by a sequence of time-boxed iterations and incremental releases. Each iteration within the Scrum framework is referred to as a sprint, which generally spans one to two weeks. For this application development, each sprint was conducted over a two-week period, including weekends. The sprint cycle consists of five key phases: Product Backlog Refinement, Sprint Planning, Implementation, Sprint Review, and Sprint Retrospective, as detailed in chapter three.

Product Backlog

User Story ID	User Story	Story Point
U001	As a user I should be able to navigate to CVDs from the homepage so that I can learn more about CVDs	16
U002	As a user I should be able to navigate to dictionary on CVDs from the homepage so that I can find meaning of words I do not understand	8
U003	As a user I should be able to navigate to newsletters and articles on CVDs from the homepage so that I can broaden my knowledge on the disease	8

U004	As a user I should be able to navigate to quiz on CVDS from the homepage so that I can test my knowledge on CVDs	8
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Table 1 Table showing product backlog and story point

Sprint One Product Backlog Refinement

T-Shirt Estimation	User Story	Justification for estimation	Need Refinement?
XL	As a user I should be able to navigate to CVDs from the homepage so that I can learn more about CVDs	This user story will need three pages	Yes
M	As a user I should be able to navigate to dictionary on CVDs from the homepage so that I can find meaning of words I do not understand	This user story will need three parts foreach word (word, pronunciation and meaning)	Yes
S	As a user I should be able to navigate to newsletters and articles on CVDs from the homepage so that I can broaden my knowledge on the disease	This user story simple and straight forward	No
S	As a user I should be able to navigate to quiz on CVDS from the homepage so that I can test my knowledge on CVDs	This user story is simple and straight forward	No

Table 2 User stories with their T-shirt Estimation

T-shirt estimation is a method used to gauge the time required to complete a user story. Each user story is categorized with a size tag ranging from XXL to S, indicating its level of complexity.

Refined User Story

User Story	Break down User Stories	Estimation
As a user I should be able to navigate to CVDs from the homepage so that I can learn more about CVDs	As a user I want to access a video animation of the types of CVDs so that it can help me understand the disease well	S
	As a user I want to access the text-based information on the type of CVD so that I can read more about the disease	S

As a user I should be able to navigate to dictionary on CVDs from the homepage so that I can find meaning of words I do not understand	As a user I should be access text-based meanings and pronunciation of the words so that I can read to understand the word	S
	As a user should be able to search for words so that I can easily look up for words	S

Table 3 Refined user stories

Final Product Backlog

Priority	User Story ID	User Story	Planning Poker Estimation
1	U001	As a user I want to access a video animation of the types of CVDs so that it can help me understand the disease well	4
2	U002	As a user I want to access the text-based information on the type of CVD so that I can read more about the disease	4
3	U003	As a user I should be access text-based meanings and pronunciation of the words so that I can read to understand the word	4
5	U004	As a user should be able to search for words so that I can easily look up for words	4
5	U005	As a user I should be able to navigate to newsletters and articles on CVDs from the homepage so that I can broaden my knowledge on the disease	8
6	U006	As a user I should be able to navigate to quiz on CVDS from the homepage so that I can test my knowledge on CVDs	8

Table 4 Final Product Backlog with priority and Planning poker estimation

Planning Poker Estimation is a valuable method for assessing the time required to complete a user story. This assessment is achieved by applying Fibonacci sequence numbers to estimate each user story. To determine the amount of work I can manage in an upcoming sprint, I utilized Estimated Velocity. The formula for calculating Estimated Velocity (EV) is as follows: Available Man-Days multiplied by the Focus Factor.

In this context, Available Man-Day refers to the number of days a single team member needs to complete a task. Since I am undertaking the sprint independently, the Available Man-Day is set at 14. The Focus Factor represents an estimation of the team's concentration level and is

calculated by dividing the Estimated Velocity from a previous sprint by the Available Man-Days. For the initial sprint, I adopted the standard Focus Factor of 0.7.

Consequently, the Estimated Velocity for the first sprint is calculated as 0.7 multiplied by 14, resulting in an EV of 9.8.

Sprint Backlog, Sprint 1

User Story	Tasks	How to demo
As a user I want to access a video animation of the types of CVDs so that it can help me understand the disease well	1. Design homepage layout using xml 2. Design the CVD page using RecyclerView to display the types of CVD 3. Create a layout for individual CVDs types in the RecyclerView 4. Create and adaptor to display the CVDs types on the RecyclerView 5. Repeat step 2 to 4 in each of the CVDs types page. 6. A frame and a video player should be inserted under the topic “overview” 7. In “res” folder, create a “raw” folder and paste the video on CVDS 8. In the CVD activity, Media Controller to play the video in the raw folder	From the homepage, click on CVDs button. Navigate to the type of CVDs you want to watch video in the CVD page and click on it. Click on the play button to play the video on the selected CVDs type.
As a user I want to access the text-based information on the type of CVD so that I can read more about the disease	1. In the layout for individual CVDs type, create TextView for topics to display on the page 2. In the “raw” folder, paste text files of the topics to display on the CVDs types page. 3. Set TextView to the text files in the raw folder using InputStream and Buffered Reader.	Navigate to the type of CVDs you wish to read on. Text based information on the CVDs type will be displayed in the page

Table 5 Sprint Backlog for Sprint One

Burn-down chart was used to monitor my daily progress. A burndown chart illustrates the amount of work completed within an epic or sprint, as well as the total work that remains. These charts help to estimate the probability of a team finishing their tasks within the set timeframe. (Rehkopf, 2019)

The daily scrum meeting was done every morning to outline the task I would complete on that day. It was held every morning from 6:00am to 6:30am. A summary of app development within the first week is as follows:

- User story U001 started with creating homepage layout. The layout was written in xml. The homepage has four clickable “cardviews”. Each cardview represent an operation of the application. The four main operation of application represented by each cardview are as follows:
- Cardiovascular Diseases (CVDs)- if selected will take user to the page showing all the types of CVDs.
- Dictionary- if selected will take users to the page where medical words related to CVDs are given explanation with an audio pronunciation of the word.
- Newsletter- if selected will take users to the page that have articles are news on CVDs. Each topic has a clickable function that takes readers to the source website.
- Quiz- if selected take users to a page that has true or false questions on CVDs.

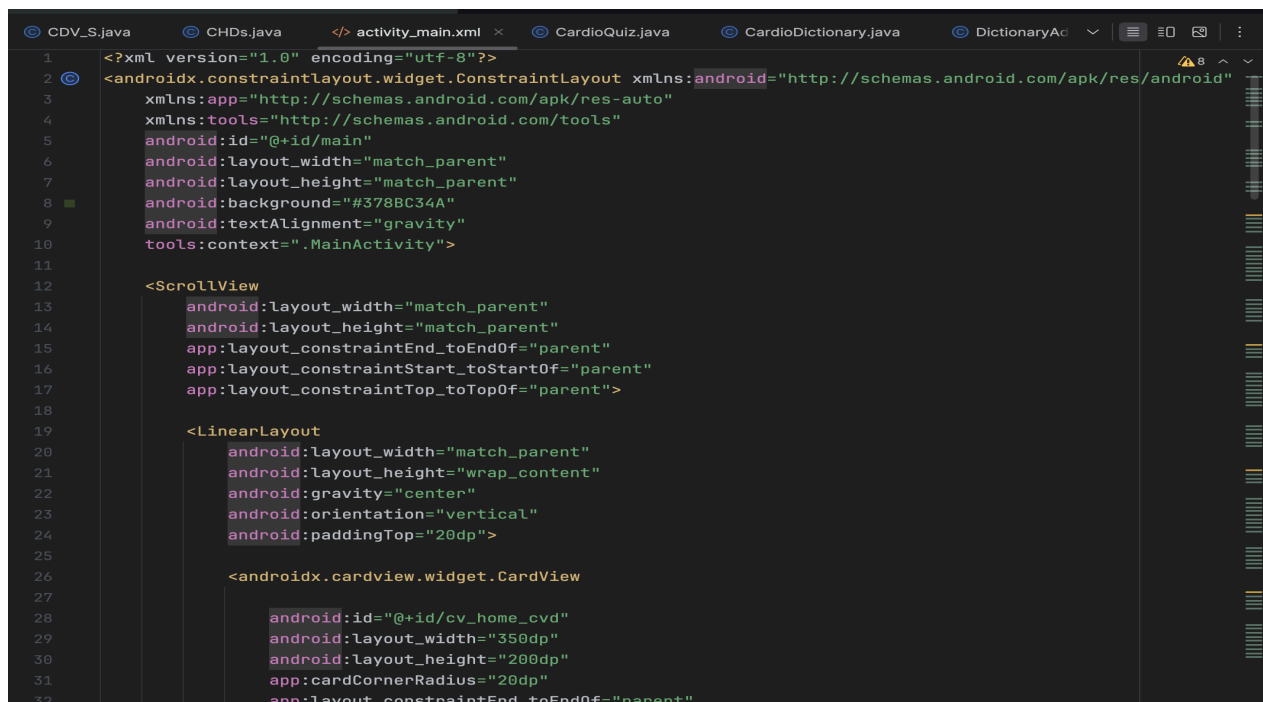


Figure 12 Part of xml code snippet of the home page

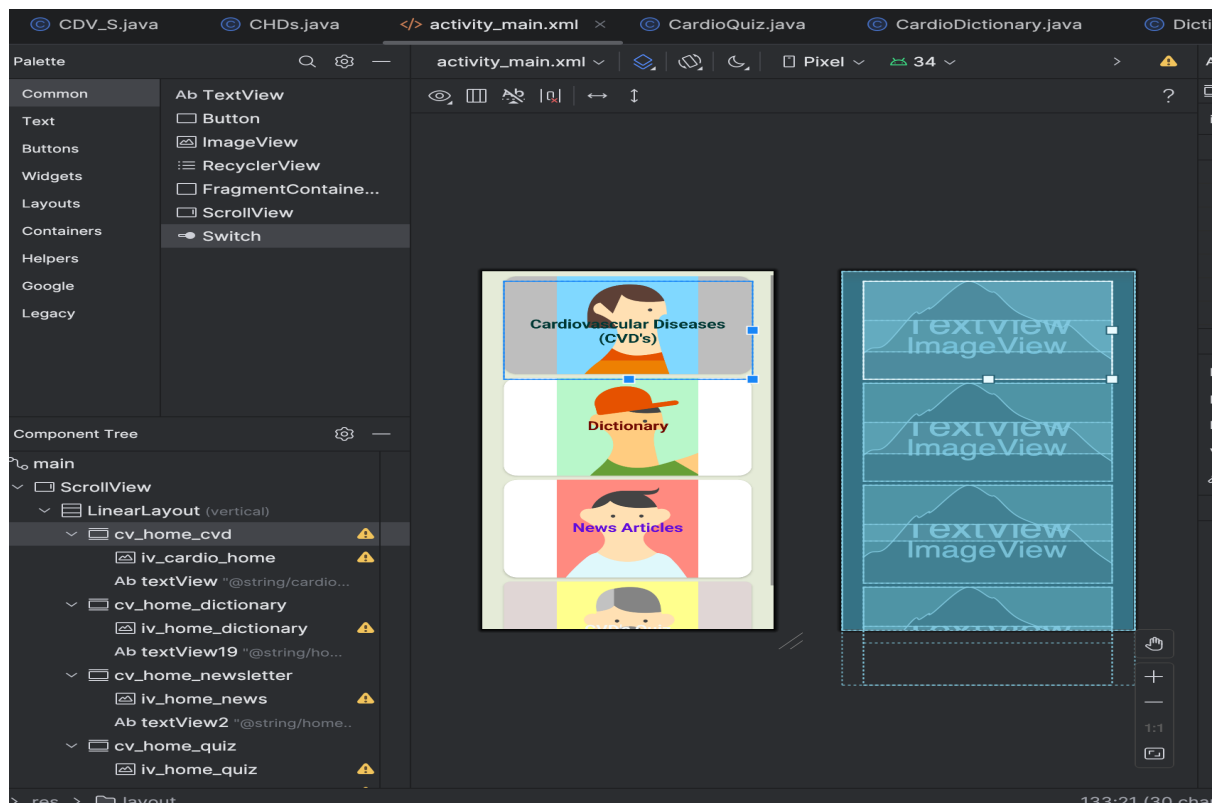


Figure 13 The homepage layout from the design view

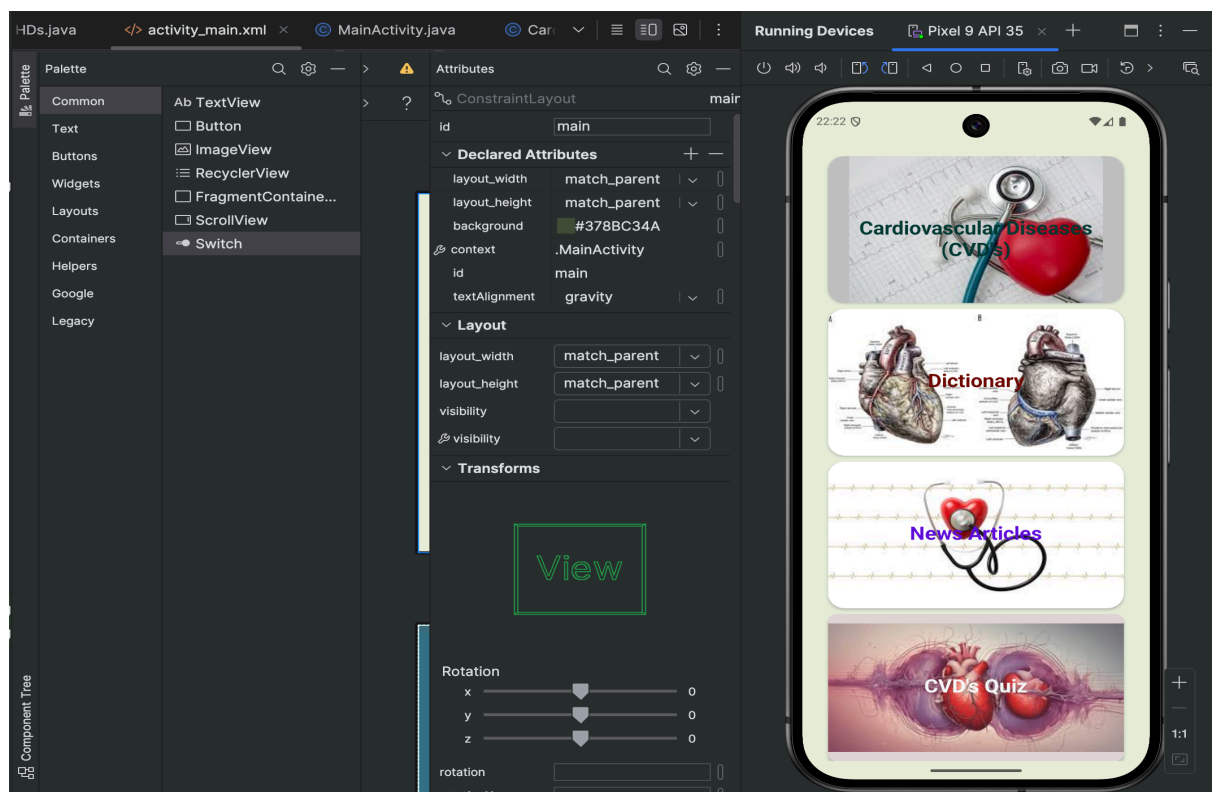


Figure 14 The application home from the emulator view

The pictures used for the cardview were not save in a folder in Android Studio. Media manager and image loading framework called Glide was used in the “main activity” to load these images from the internet.

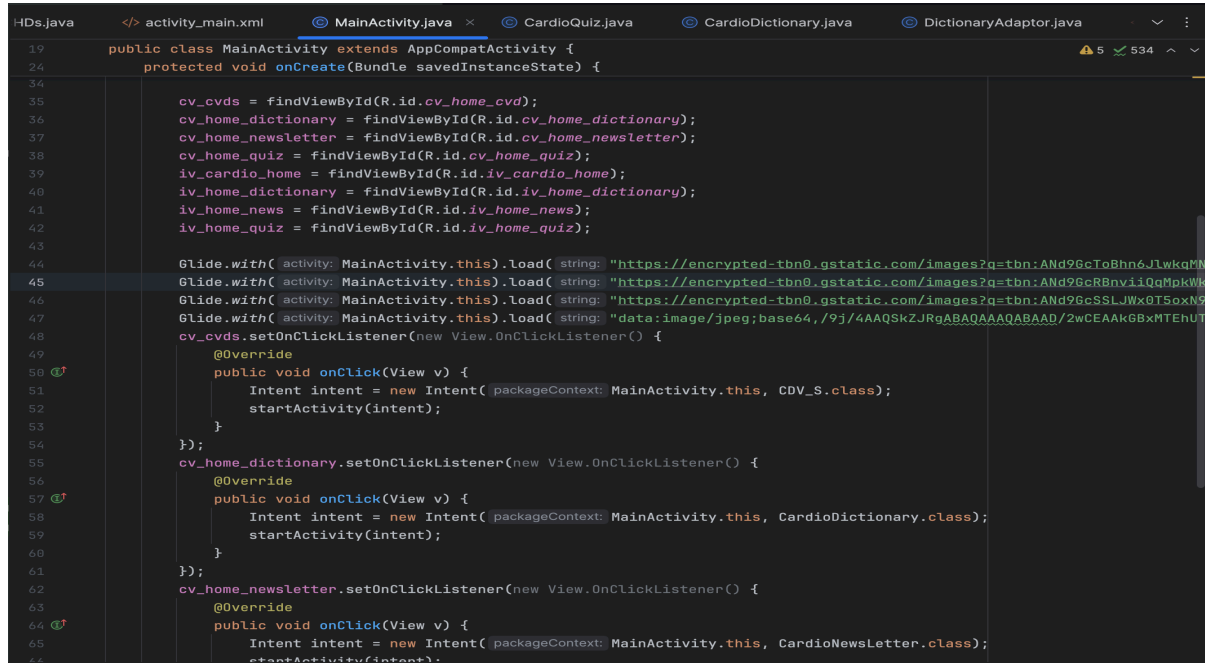


Figure 15 Code snippet from the main activity showing image loading with Glide and the clickable functions of the cardviews

The Cardiovascular (CVDs) button when clicked will open the CVDs page displaying the types of CVDs. Each type is a button when clicked opens the page for the selected type. The information on each type is organised under the following topics:

- Overview: this has a video animation on the selected type of CVDs and a brief description on the diseases in text.
- Signs and Symptoms: this displays the symptoms of the disease
- Causes: causes of the disease is displayed here
- Diagnosis: this has the medical diagnosis of the diseases
- Management: how the disease is managed medically and lifestyle changes to avoid it.

The CVDs layout used recyclerview to showcase the types of CVDs. Two “classes” and a layout were created for this, namely, the Adaptor Class and CreateModel Class. The CreateModel class was used to create the type of CVDs, while a layout was design to display a single item within the recyclerview.

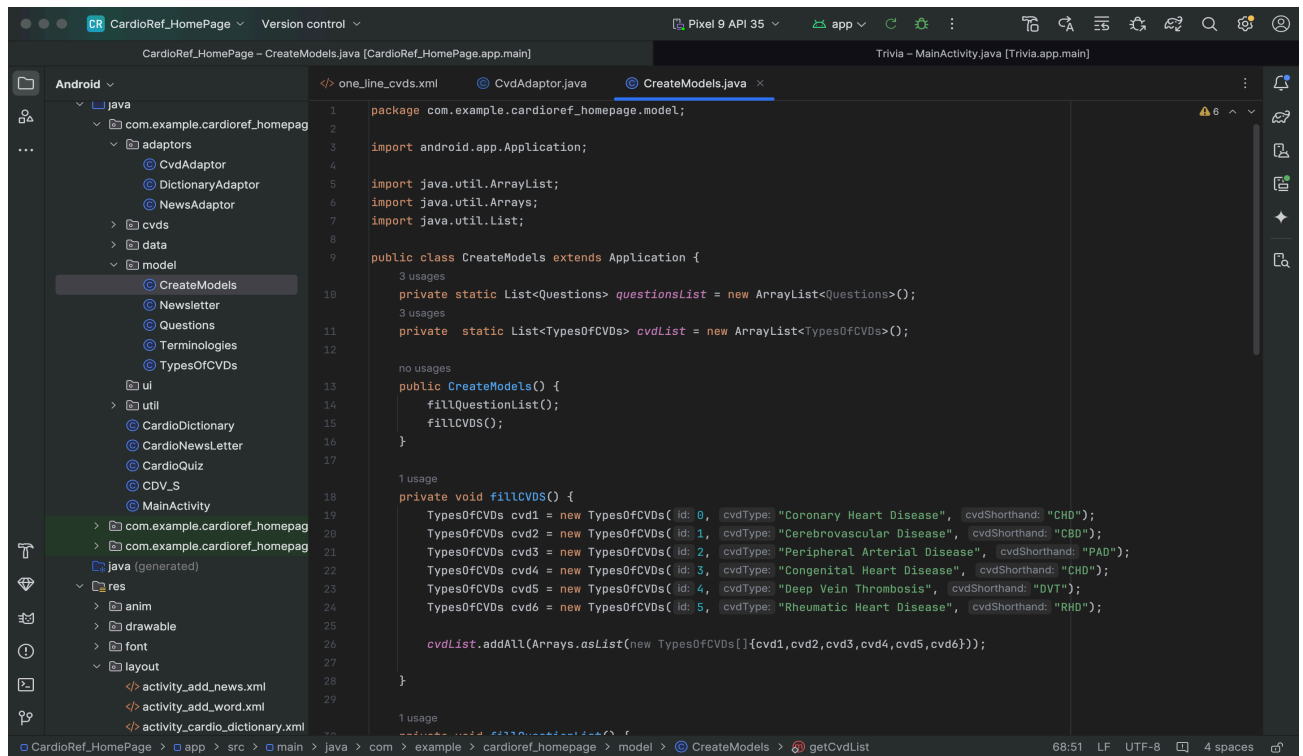


Figure 16 A screenshot of the CreateModel Class that was used to create the types of CVDs for the adaptor to display on the CVDs page

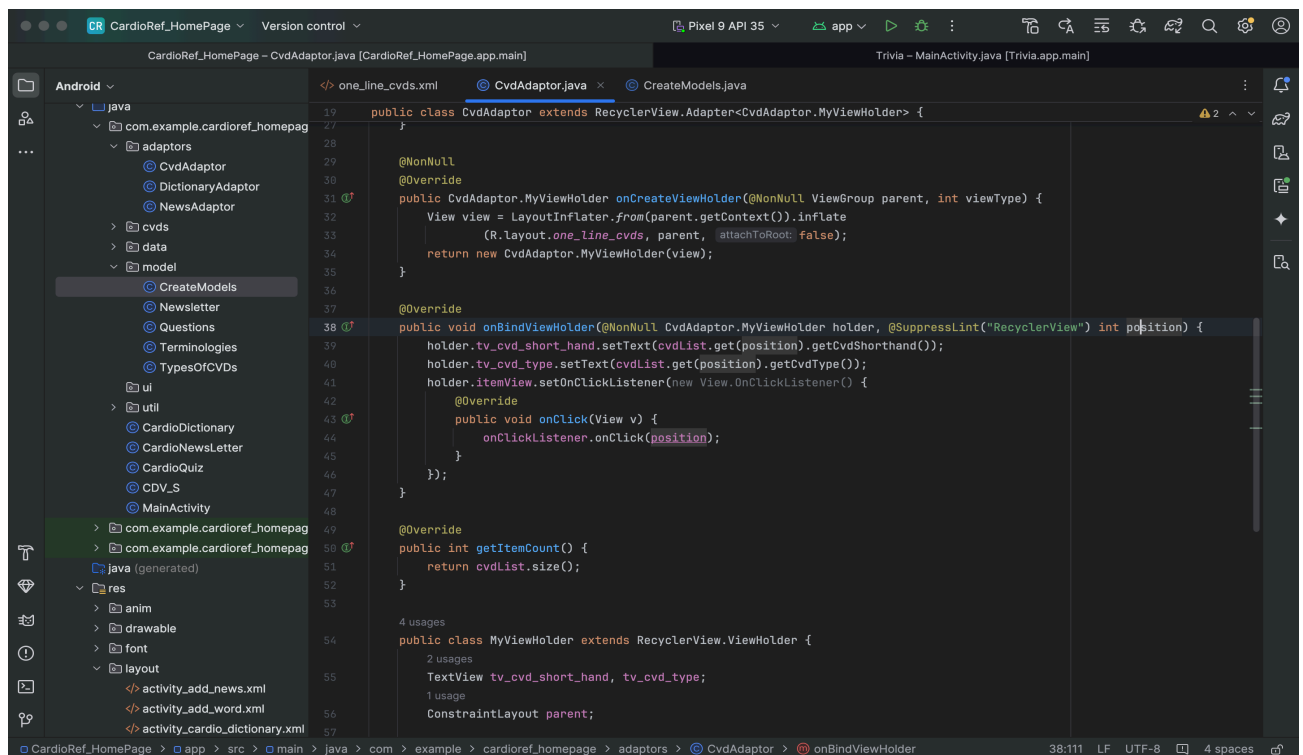


Figure 17 A screenshot of code snippet of the adaptor created for the recyclerview on CVDs page layout

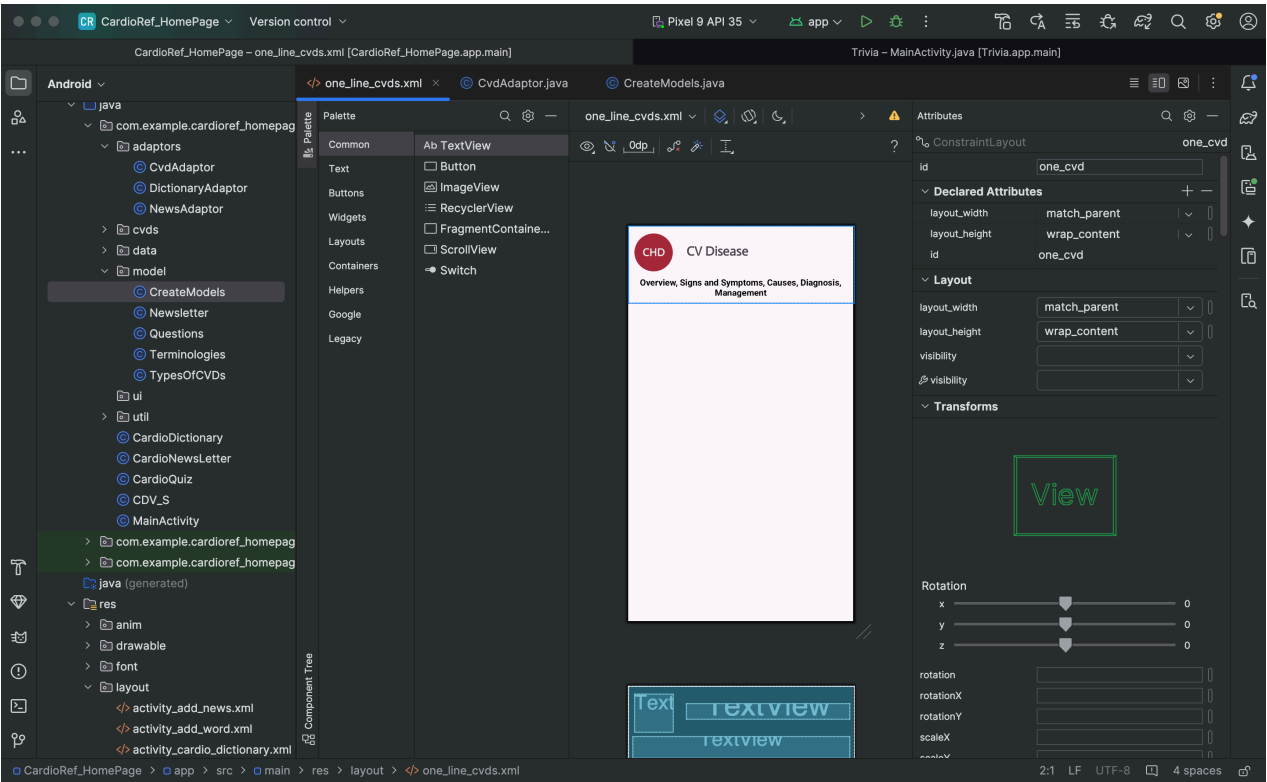


Figure 18 a picture of the layout created for individual item in the recyclerview on the CVDs page

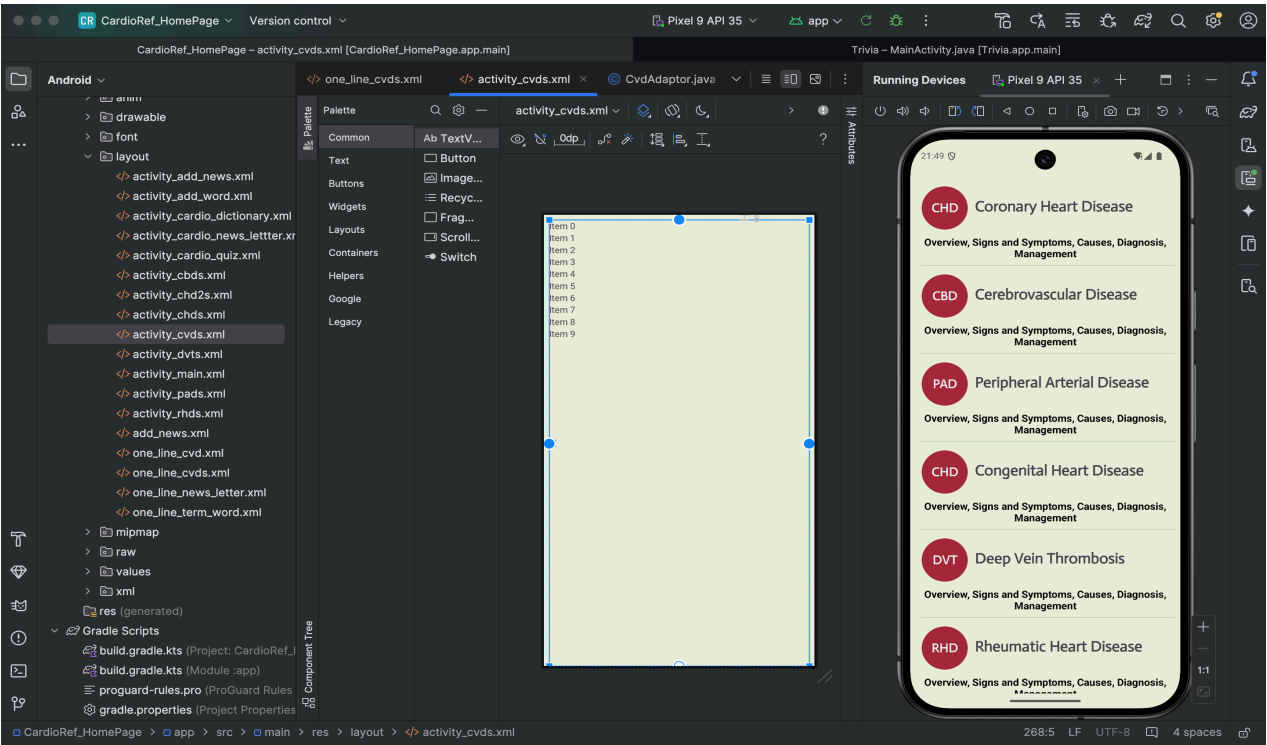


Figure 19 The emulator view after joining the classes and the layout together.

The development of user story U001 included user story U002. A “raw” folder was created in Resource folder, where the animated videos and the texts to display under the topics of each individual CVDs where stored. The texts were in “txt” format whiles the videos were in “mp4” format. All the layout for the types of CVDs was the same, therefore a single layout was design and included in each of the CVDs types activity layout.

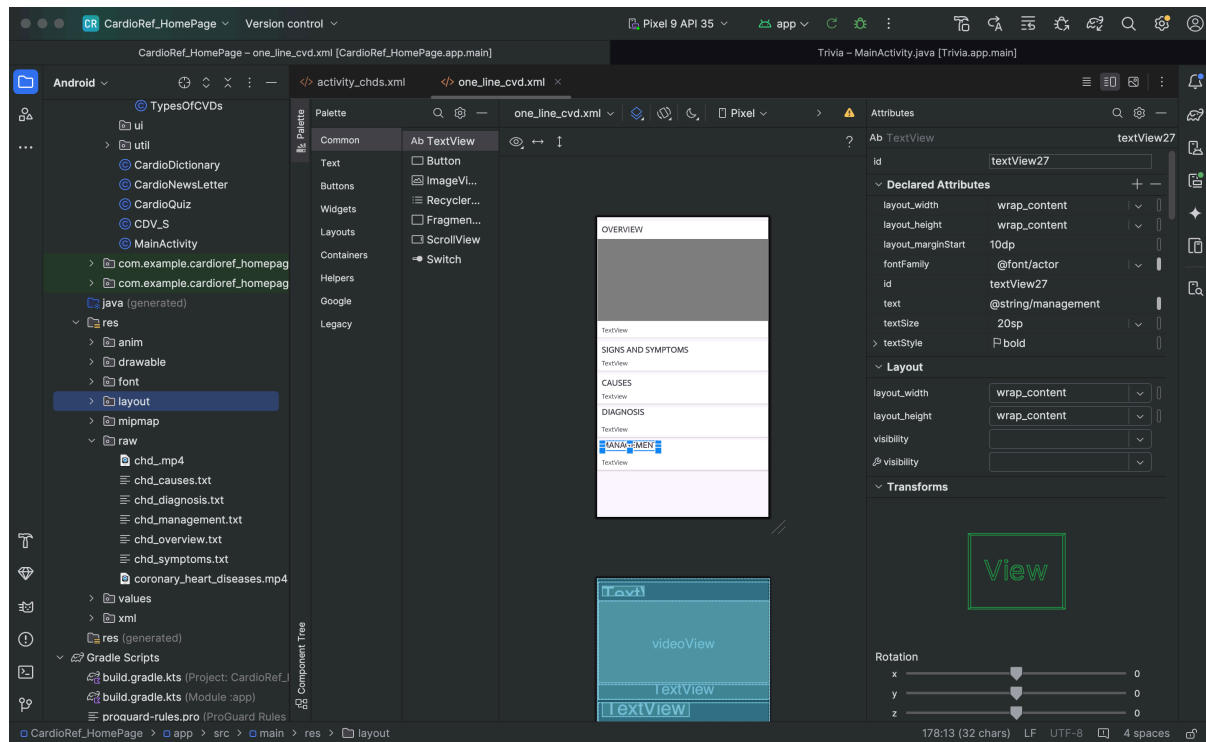


Figure 20 A picture of the layout for the type of CVDs page

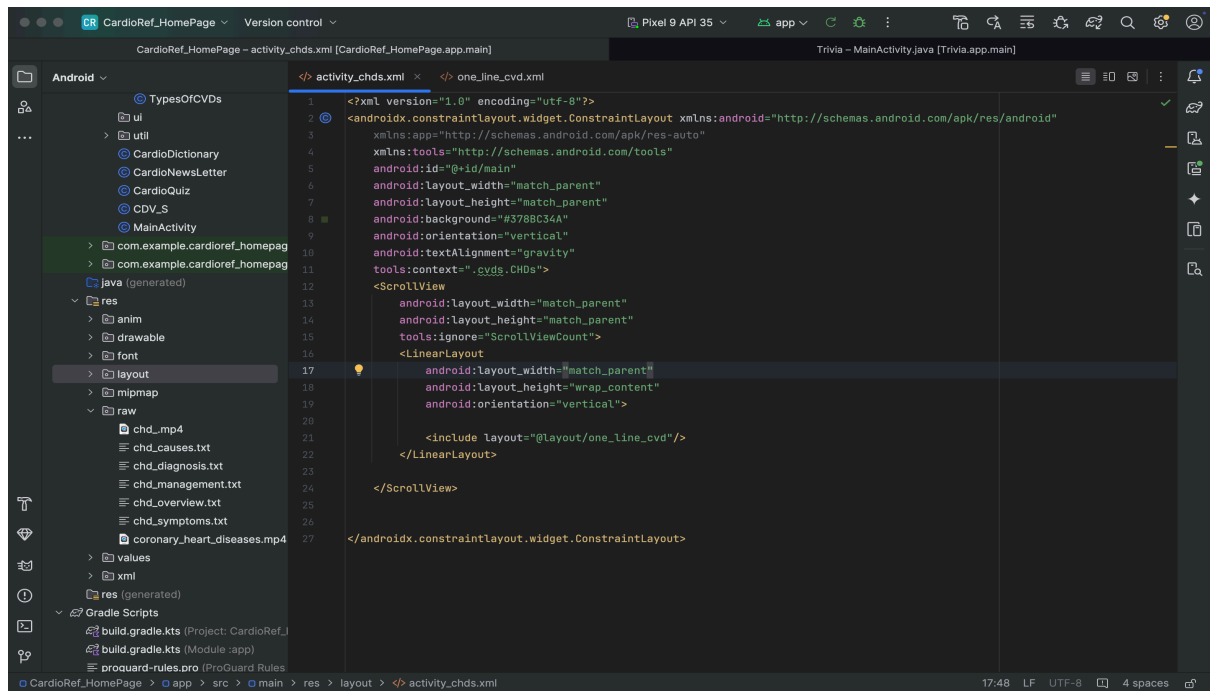


Figure 21 A picture showing the inclusion of the layout in Fig 19 in the main layout of one of the types of CVDs

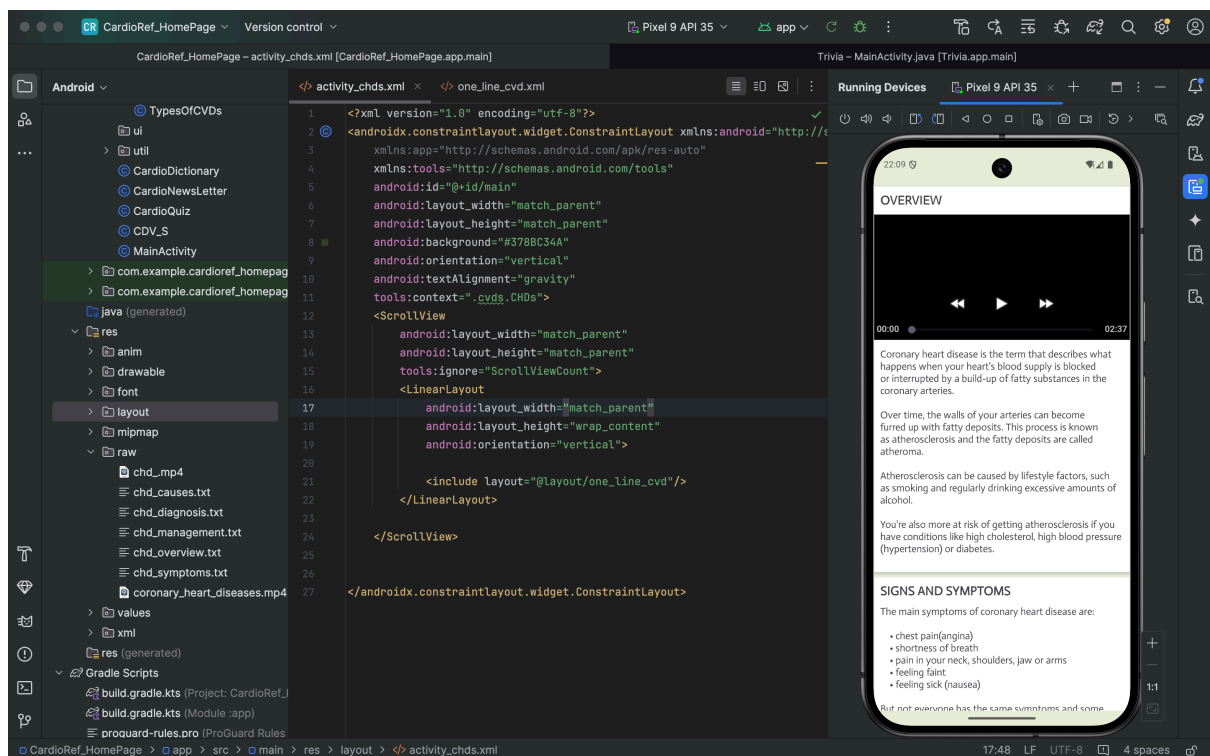


Figure 22 Emulator view of combining the files in the raw file and the layout on one of the types of CVDs page.

User story U001 and U002 were completed in the first week of development. Below is the Burn-Down chart for user story U001 and U002.

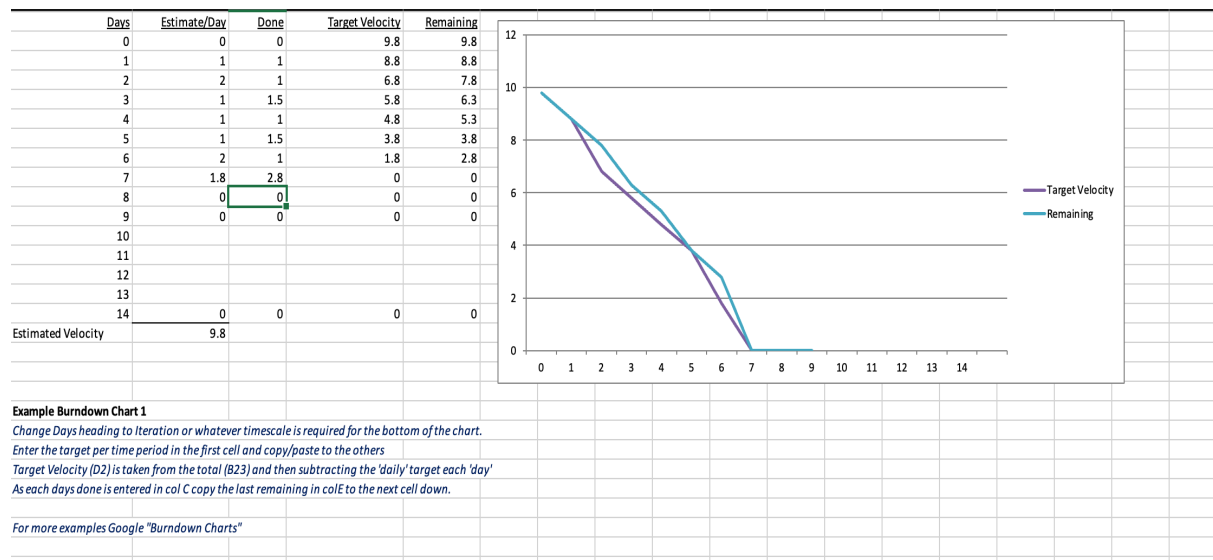


Figure 23 Picture of Burn-Down Chart for User Story U001 and U002

In week two in Sprint One, user story U003 and U004 was added to the Sprint. The two user stories make up the Dictionary function of the application. Layout design for the page would include the two user stories.

User Story	Task	How to demo
As a user I should be access text-based meanings and pronunciation of the words so that I can read to understand the word	• Create a class to instantiate the dictionary word/terminology objects • Create SQL Lite database for the dictionary words/terminologies. • Create schema for the database • Create a page to perform CRUD functions to the database. • Implement recyclerview in the dictionary page layout to display the words, their meaning and pronunciation from the database • Create a layout for a single item in the recycler view • Create adaptor to implement the recyclerview on the dictionary page using the layout created for a single item.	Click on the Dictionary button on the homepage. Select a work/terminology to hear its pronunciation and read the meaning of the word/terminology.
As a user should be able to search for words so	• Add searchview to the dictionary layout.	Navigate to the search view. Type in the

that I can easily look up for words	Implement the search function in the dictionary activity	words/ terminology to filter from the list of words/terminologies
--	--	---

Table 6 Table showing the additional user stories added to the first sprint in the second week of development

In the second week of the first sprint, the Dictionary feature of the application was developed. As shown in Table 6, the Dictionary feature had two user stories, and as they go hand in hand, development of these two user stories were developed hand in hand. The development followed the tasks in the in Table 6. A “Terminology” class was created to serve as the blueprint for the Dictionary objects as shown in Fig 23. The parameters declared in this class was used to create the schema for the database.

```

4
27 usages
5 public class Terminologies {
6     4 usages
7     private int id;
8     4 usages
9     private String terminology;
10    4 usages
11    private String word_audio;
12    4 usages
13    private String meaning;
14
15    1 usage
16    public Terminologies() {
17    }
18
19    2 usages
20    public Terminologies(int id, String terminology, String word_audio, String meaning) {
21        this.id = id;
22        this.terminology = terminology;
23        this.word_audio = word_audio;
24        this.meaning = meaning;
25    }
26
27    public int getId() {
28        return id;
29    }
30
31    public void setId(int id) {
32        this.id = id;
33    }
34
35    5 usages
36    public String getTerminology() {

```

Figure 24 A picture of the Terminology class created for the Dictionary objects

A database handler class was created for the database. This class extends the “SQLiteOpenHelper” class. SQLiteOpenHelper is a helper class design to facilitate the creation and management of database versions. Inside this class, the Terminology table was created as well as methods to perform CRUD operations on the database. The schema for this database were the parameters declared in the Terminology class.

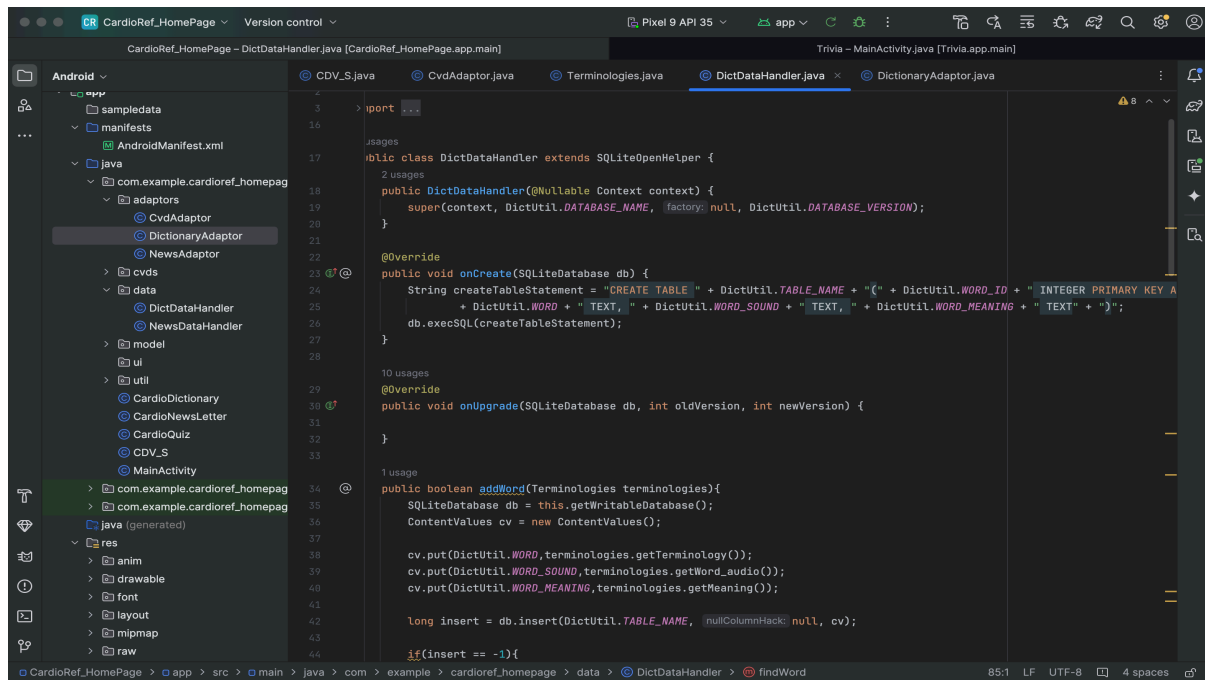


Figure 25 Code snippet of database handler user to create the table for the Dictionary/Terminology

To enable me manage data in the database table, I created a page, hidden from the user, which allow me to perform CRUD operations on the database table. Fig 25 shows a picture of the SQL Lite database in a SQL Lite application after words have been add to the database table.

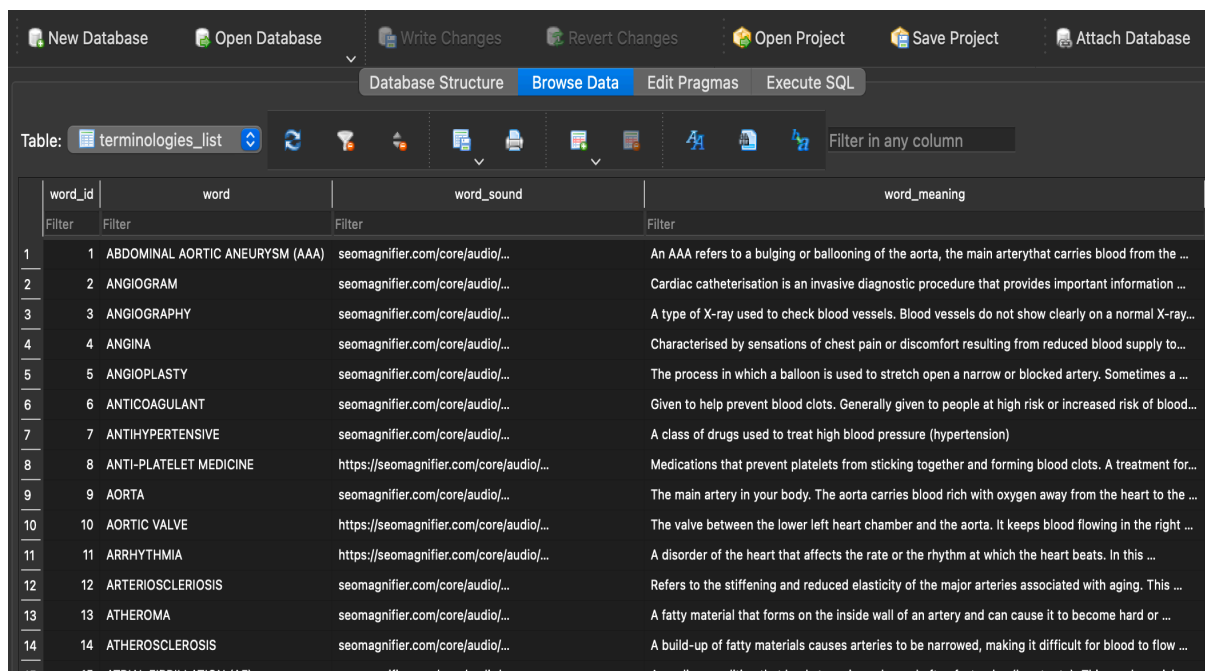


Figure 26 A picture of SQL Lite application showing the data added to the Terminology table.

The Dictionary activity layout used recyclerview to display data from the database. Therefore, a layout for a single item on the recyclerview and an adaptor was created for the Dictionary page. The single item layout displays the Terminology/word, play the pronunciation of the word and the meaning of the word. The adapter used MediaPlayer class to play words pronunciation in mp3 format from the internet. Word search was implement using searchview in the Dictionary activity. Fig 25 shows the dictionary from the emulator after completing all the tasks in user story U003 and U004.

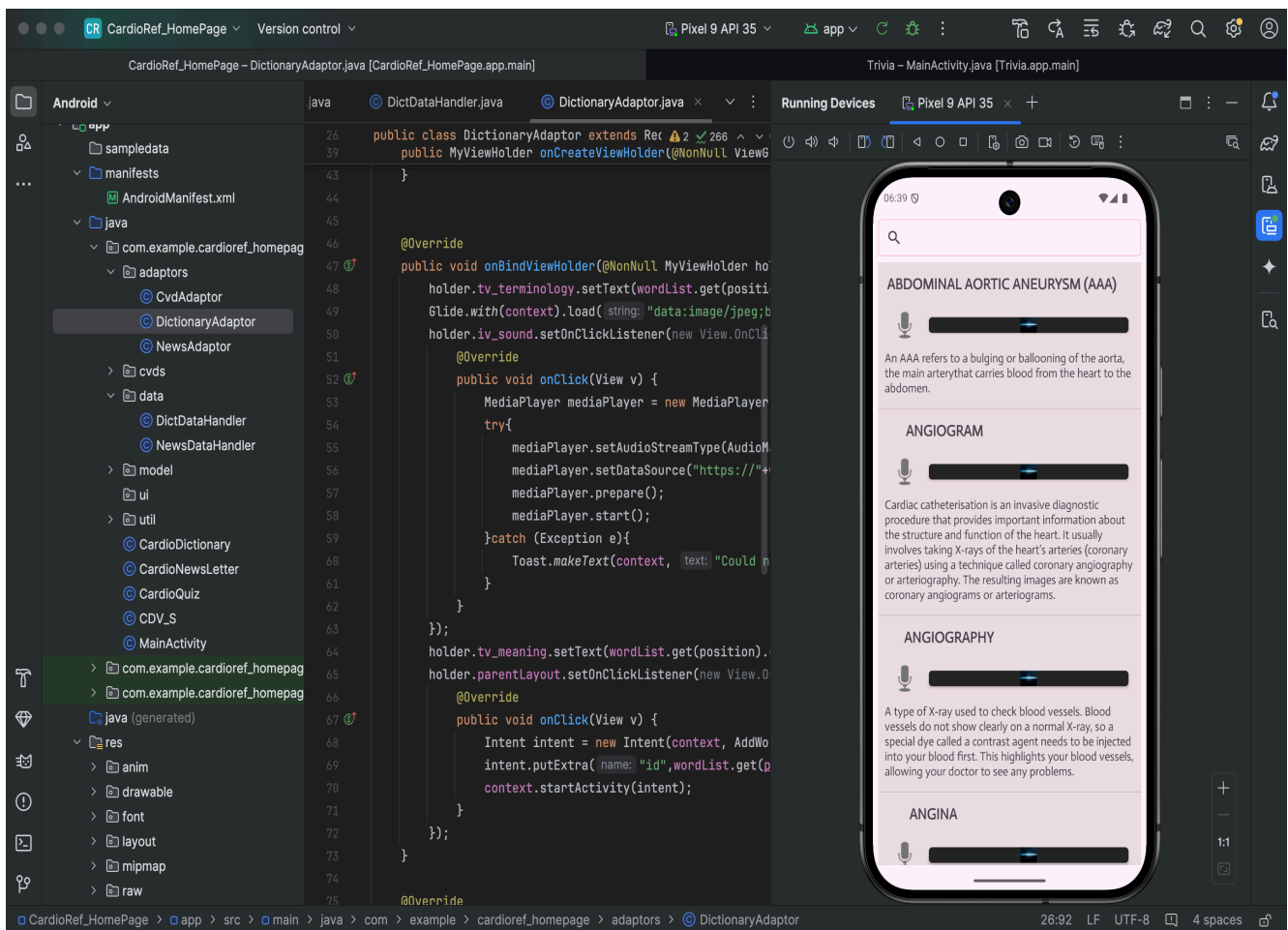


Figure 27 A screenshot showing the emulator view of the Dictionary page after development

Sprint One Reflection

The Burn-Down chart for the first sprint is shown below.

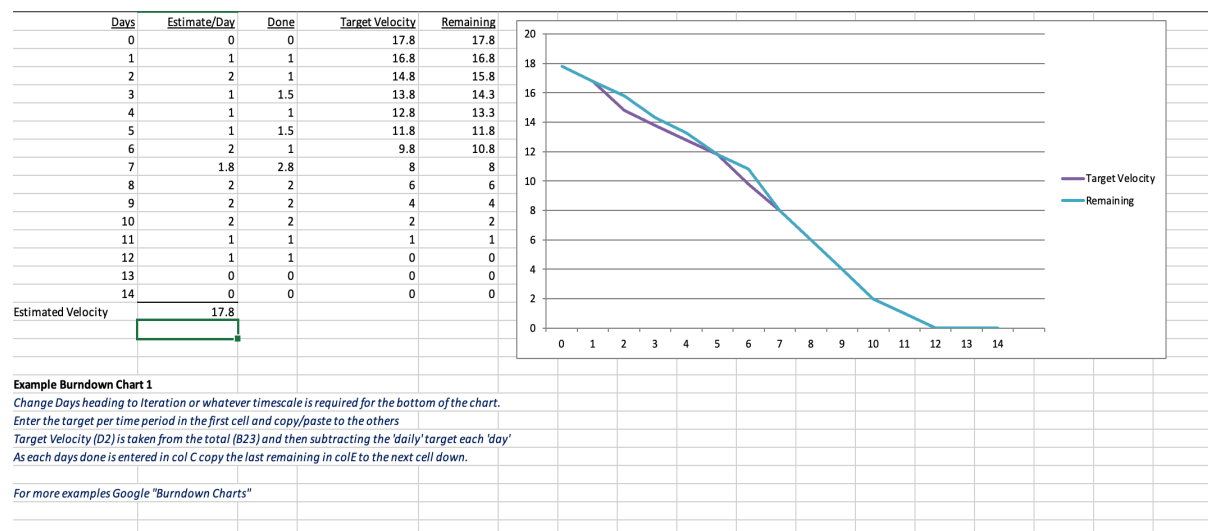


Figure 28 A picture showing the Burn-Down chart for Sprint One

The Burn-Down chart indicates that the application's development is progressing more rapidly than I had originally anticipated. This acceleration can be attributed to the availability of APIs for Android, which facilitate a more efficient and streamlined development process.

While there were several risks identified, the most significant one was the potential interruption of development due to illness. Given that I am the sole developer on this project, such an event could have a substantial impact. To mitigate this risk, I have resolved to maximize my daily productivity, often surpassing the goals I establish for myself each morning prior to commencing work. This risk has been acknowledged, and appropriate measures have been taken in my subsequent sprints.

Sprint One Retrospective

Date: 21/1/2025

Duration: 14 days

Sprint Goal: Kickstart the development of the CardioRef mobile application, focusing on core features like CVDs features, basic user interface design, and integrating initial educational content.

What Went Well:

- **Initial Planning and Setup:** I did a great job of setting clear goals and outlining the scope for the first sprint. I defined the key features (CVDs, CVD dictionary, Newsletter and Quiz) and prioritized them effectively.
- **Basic UI Implementation:** The user interface (UI) design was successfully initiated, with basic navigation and layout set up for the CVDs and content sections.
- **Content Development:** The initial educational content about CVD was collected from trusted sources (WHO, ESC) and formatted for the app. The glossary of CVD terminology was created and integrated into the app.

What Could Be Improved:

- **Content Integration:** The process of formatting and integrating educational content proved to be more time-consuming than initially planned, which delayed the completion of some sections of the app.
- **User Testing:** I didn't engage with external testers during this sprint. As a result, some assumptions about user experience were not validated early enough, which could have helped refine the design.

What I'll Do Differently Next Sprint

- **Stay Focused on Core Features:** We need to stick more closely to the original scope for the next sprint, focusing on perfecting the essential features (Newsletter and Quiz features)
- **Streamline Content Integration:** Set up a more efficient process for content integration by defining standardized formats and templates for educational materials, reducing formatting time in future sprints.
- **Early User Feedback:** Start engaging with a small group of target users early to get feedback on the app's usability and educational content, which will help me adjust sooner rather than later.

Overall Reflections

The first sprint set a strong foundation for the CardioRef project. I successfully established the basic framework of the app, developed the CVDs and Dictionary feature. However, there were some learning curves regarding the development process and content integration that slowed progress. Moving forward, we'll focus on streamlining content integration, ensuring better time management, and engaging users earlier to get actionable feedback on design and functionality.

Sprint Two

The Actual Velocity for Sprint One is recorded at 17.8, with a total of 14 Man Days allocated. Consequently, for Sprint Two, the Focal Factor is calculated as the Actual Velocity of Sprint One divided by the Man Days, resulting in $17.8/14 = 1.27$. Utilizing this data, the Estimated Velocity for Sprint Two can be determined by multiplying the Man Days for Sprint Two by the Focal Factor, yielding $14 * 1.27 = 17.8$. The sprint backlog for Sprint Two is detailed in the table below, derived from the product backlog.

User Story	Task	How to Demo
As a user I should be able to navigate to newsletters and articles on CVDs from the homepage so that I can broaden my knowledge on the disease	• Create a class to instantiate news/article objects • Create SQL Lite database for the news object • Create schema for the database • Create a page to perform CRUD functions to the database. • Implement recyclerview in the Newsletter page layout to display news/article headline, source, dated posted/updated and am news/article image from the database • Create a layout for a single item in the recycler view • Create adaptor to implement the recyclerview on the Newsletter page using the layout created for a single item.	Navigate to Newsletter page. Scan through the articles posted on the page. Select an article to be directed to the original source via a default browser
As a user I should be able to navigate to quiz on CVDS from the homepage so that I can test my knowledge on CVDs	• Create a class to instantiate Quiz true/false questions objects • Create the Quiz objects in the CreateModel folder • Create anim folder the res folder for animation on the Quiz page.	Navigate to Quiz page. Answer the question either by choosing true or false. Depending on the answer, an animation should show. Click on the next button or previous

	• Display the questions on the Quiz page with true/false buttons to select the answer. • The page should show an animation depending on whether the answer is wrong or correct. • Previous and next button on the page to move questions forward or backwards	button to navigate question forward or backwards.
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Table 7 Table showing the sprint backlog for Sprint Two.

Deliverables in Sprint Two follows the tasks in the sprint backlog table. User Story U005 started with creating a Newsletter Class. it creates a newsletter with news headline, date posted/updated, source of news/article as parameters. This user story development follows the same steps in the development of User Story U003 and U004. A database was created using SQL Lite. Schema for the database table was created from the Newsletter Class parameters. Recyclerview was implemented in the Newsletter page and a layout for a single item in the recyclerview was created as well. Data was retrieved from the database and displayed on the Newsletter page using a recyclerview adaptor.

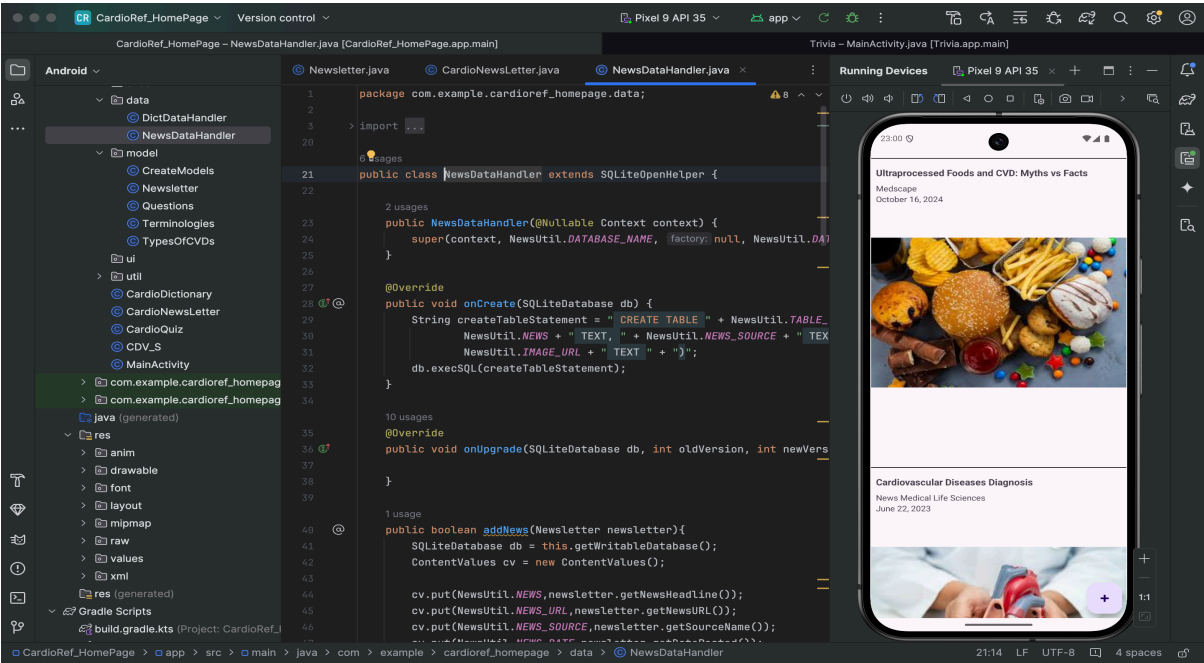


Figure 29 Emulator view of completing User story U005

If an item on the recyclerview is clicked, the source website of the news or article is open via a browser. These tasks were pretty much completed on a faster side since most of the structures have been implemented on other pages.

The last feature of the Application, started with creating a Questions Class to instantiate question object. Each question object has an integer ID, a question in string literal and a Boolean to determine whether the question is true or false. The question object is created in the CreateModels Class and displayed on the Quiz page.

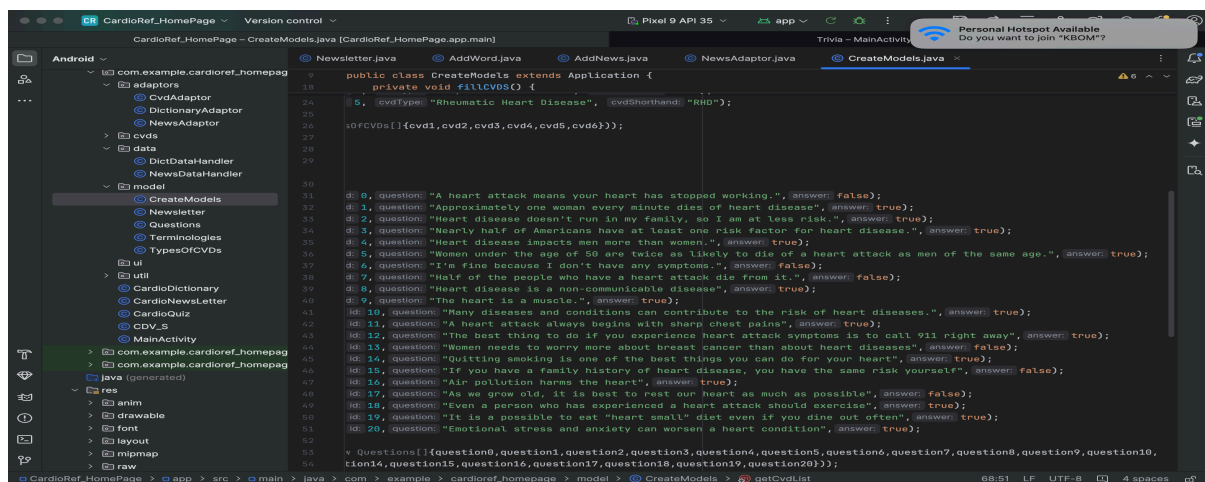


Figure 30 A picture of a code snippet showing how the questions were created in the CreateModels Class

The Quiz page had a single layout with Textview on a Cardview. Questions were displayed on the Cardview via the Textview. The page also has a next button and the previous button to navigate questions to the next question or previous question.

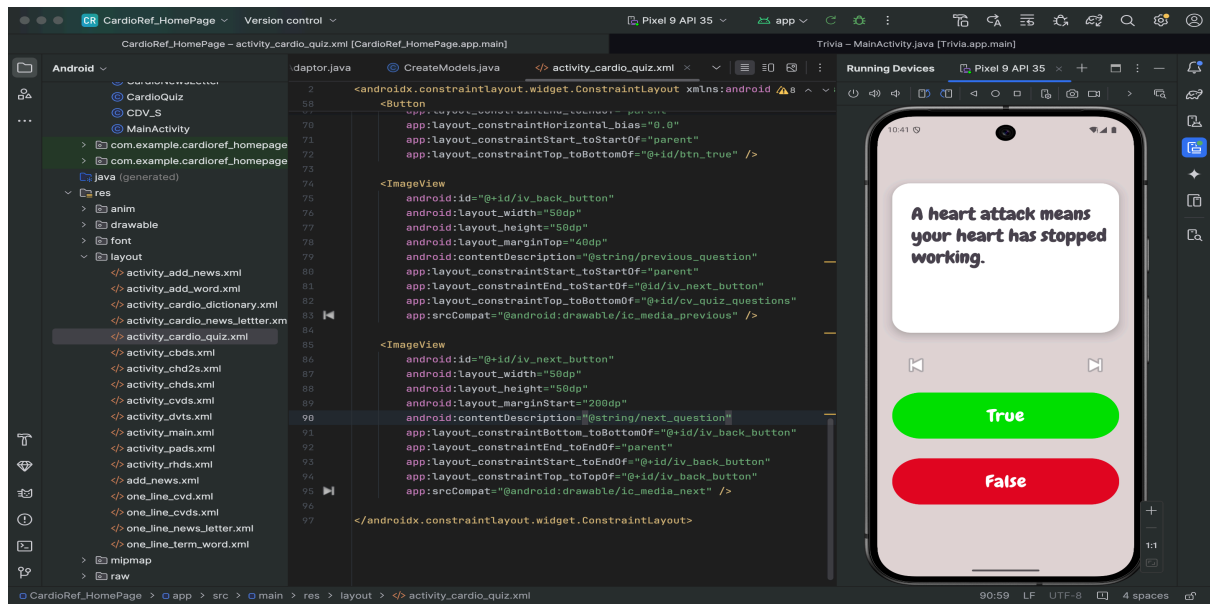


Figure 31 A screenshot of the Quiz page from the emulator view

The page implemented an animation from the “anim” folder where the Cardview shakes and changes color to either red or green color depending on whether the answer clicked is correct or not. This shake animation is accompanied by a Toast message to show whether a chosen answer is correct or incorrect.



Figure 32
is chosen

A screenshot showing the animation and Toast message when a correct answer

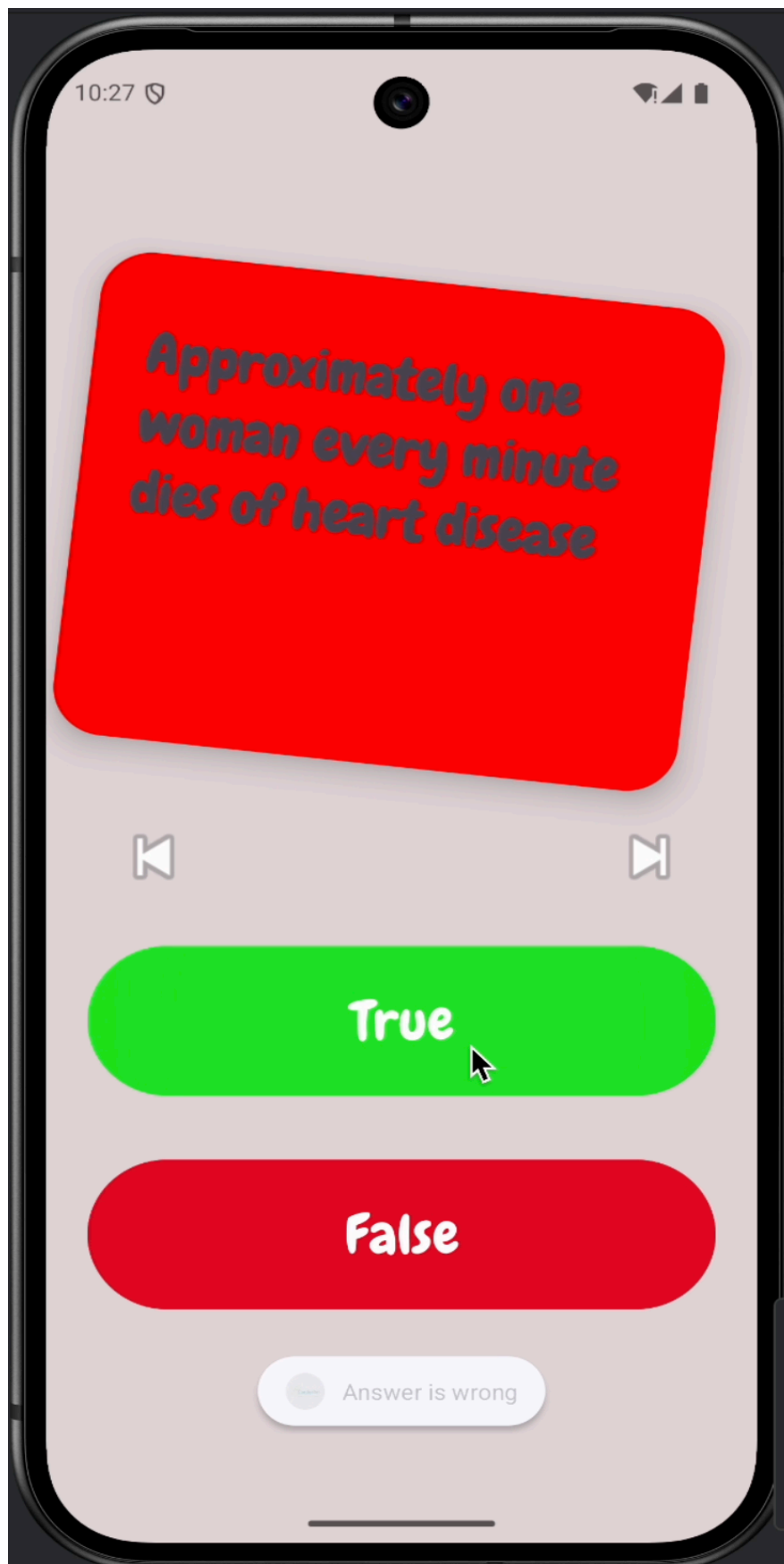


Figure 33 A screenshot showing the animation and Toast message when a wrong answer is chosen

The whole application development ended in two Sprints. The duration for the two sprints in 28 days. Approximately, 8 hours a day was spent a day during the development stage.

Sprint Two Review

The progress of Sprint Two is illustrated in the Burn-Down chart presented in Fig 34. The risk management strategies employed during this sprint facilitated a seamless development process. This sprint spanned a duration of 14 days, with an average of five hours dedicated to the development phase.

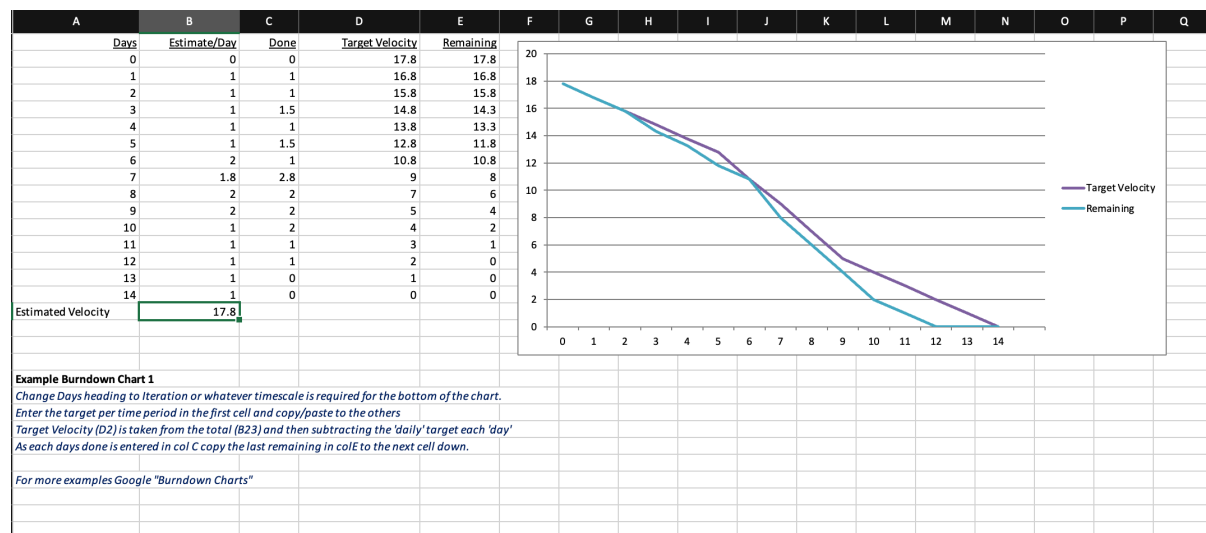


Figure 34 Burn-Down chart for Sprint Two

6 Testing and Evaluation

I conducted end-user functionality testing following the completion of this application. This phase in software development involves users evaluating the application's features and functions in real-world scenarios to ensure it aligns with their needs and expectations. This process verifies that the software performs correctly from the users' viewpoint. A total of 15 participants engaged in this testing. The tests focused on four primary functionalities of the application, which include:

- Obtaining information about various types of CVDs
- Utilizing the Dictionary to understand terms related to CVDs
- Accessing articles and news through the application
- Assessing their knowledge of CVDs via a true or false CVD game.

The table below presents the testing results, where "Success Guarantee" indicates that users successfully executed a functionality and achieved their objectives, while "Minimum Guarantee" signifies that users were unable to accomplish their goals due to unsuccessful execution of a function.

Function	Goal	Success Guarantee	Minimum Guarantee
Getting information on the types of CVDS	Navigate to a type of CVD, play the video on the CVD type and read the explained text information on the page	15	
Using the Dictionary to find meaning of words relating to CVDs	Navigate to the Dictionary Page, use the search engine to search word, play the pronunciation of the word and get explained text on the word	15	
Getting articles and new from the application	Navigate to the Newsletter page, select an article or news, click on it to be directed to the article or news source website to get full information	15	
Testing their knowledge on CVDS through a false or false CVD game.	Navigate to the Quiz page to answer True or false questions on CVDs	15	

Table 8 Table showing results of end-user functionality test

The achievement was accompanied by valuable feedback from users. Below are some of the recommendations received:

- Users suggested omitting the dictionary feature and instead providing a link on the CVDs type page that directs them to a website offering articles or definitions for the words.
- The quiz should incorporate additional features, such as displaying the highest score, including an animation when a user achieves a new high score, and allowing users to retake the quiz after answering the final question.

To gather participants' opinions on the application, a brief questionnaire was created. This questionnaire consisted of three sections: appearance and design, UI testing, and user experience, with each section containing three questions.

CardioRef is an android application developed to give users information on cardiovascular disease or heart diseases. The Goal of the application is to create awareness on the disease and how to prevent it. As part of the user testing experience, participants are required to answer these questions. This questionnaire is anonymous and does not require user personal information. Feedback from this questionnaire will help in improving the application in the future.									
Appearance and Design choose from 1- strongly disagree, 2 - disagree, 3 - not sure, 4 - agree 5 - strongly agree - The application has a good design 1 2 3 4 5 - The colours used is acceptable 1 2 3 4 5 - The appearance of the app is acceptable 1 2 3 4 5									
UI Testing choose from 1- strongly disagree, 2 - disagree, 3 - not sure, 4 - agree 5 - strongly agree - The application was easy to navigate to desired pages 1 2 3 4 5 - The UI is well structured 1 2 3 4 5 - The UI is user friendly 1 2 3 4 5									
User Experience choose from 1- strongly disagree, 2 - disagree, 3 - not sure, 4 - agree 5 - strongly agree - The application provided good information 1 2 3 4 5 - The application is easy to use 1 2 3 4 5 - The application was able to achieve its goal 1 2 3 4 5 What your favourite function of the application 1-CVDs 2-Dictionary 3-Newletter 4-Quiz									

Figure 35 A prototype of questionnaires used along with functional end-user testing

The survey data was analyzed utilizing Microsoft Excel. The findings were categorized into three subtopics, with a pie chart employed to visually depict the distribution of the data.

In the category of Appearance and Design, 12 participants indicated that the application was user-friendly, allowing them to easily access their desired pages, which accounts for 80% of the total responses. Additionally, 3 participants expressed strong agreement, representing 20% of

the total responses. The results were consistent with those of the other two questions within the Appearance and Design category.

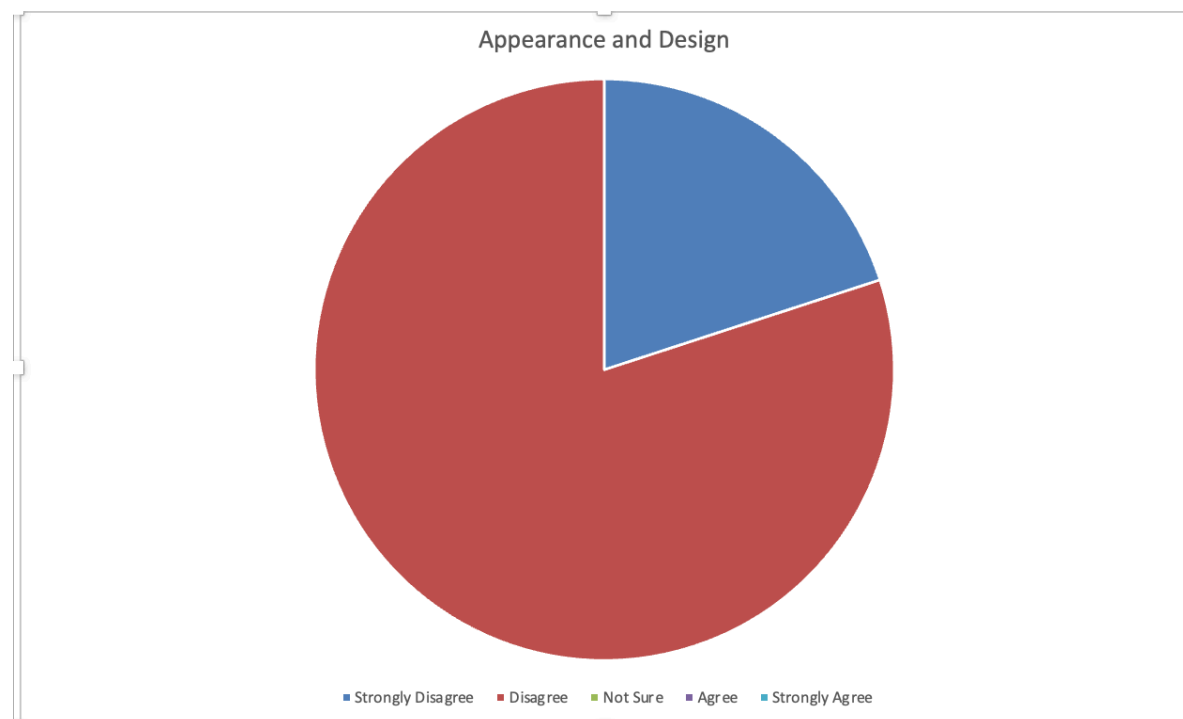


Figure 36 A pie chart showing data distribution under Appearance and Design

In the UI Testing, 10 participants responded affirmatively to the three questions, accounting for 66.7% of the total responses. One participant expressed uncertainty, representing 6.7%, while the remaining two participants, making up 26.6% of the responses, indicated strong agreement.

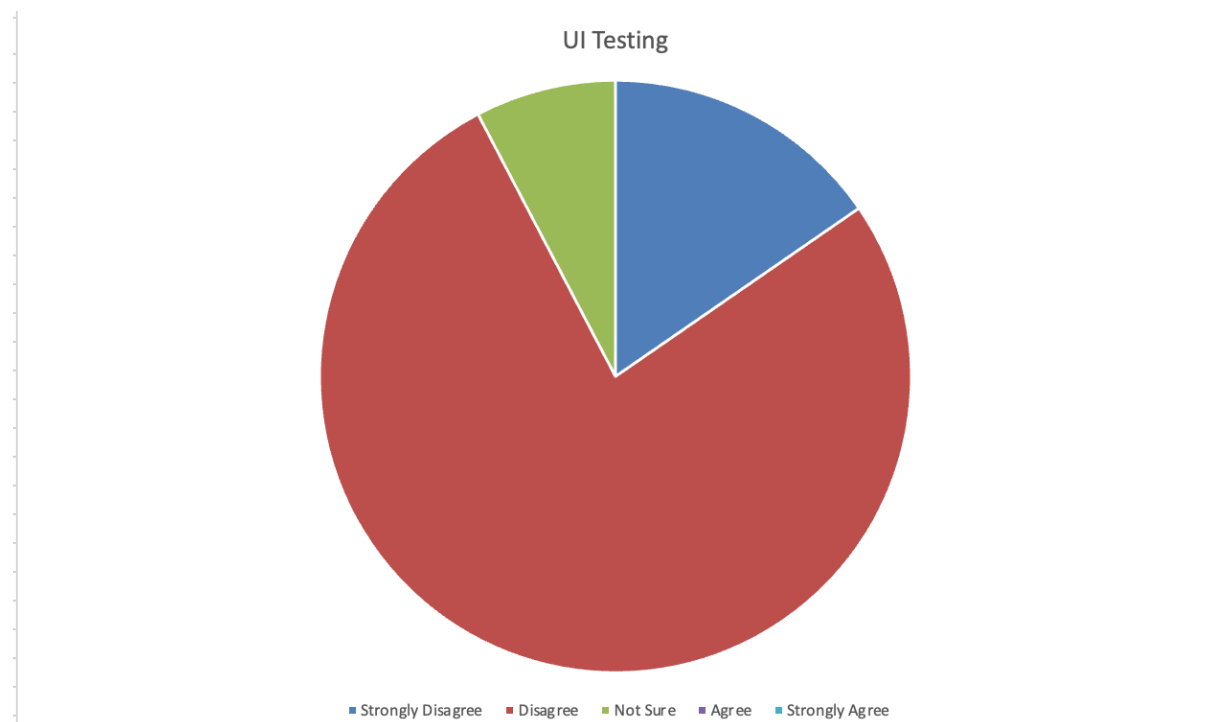


Figure 37 Pie chart showing the data distribution of UI Testing

User Experience yielded impressive results, with 14 out of 15 participants affirming their agreement to the three questions posed, which accounts for 93.3% of the data frequency; the lone participant expressed uncertainty. The quiz emerged as the most favored feature, chosen by 9 participants, while 5 preferred the CVDs feature and 2 opted for the Newsletter.

The outcomes from the Testing and Evaluation were highly positive, reflecting the participants' feedback. This provided valuable insights into how to effectively use language to enhance awareness of CVDs through the application. It became evident that additional features could be integrated into the application, not only to raise awareness but also to facilitate the monitoring of users' cardiac health. Incorporating customization options and user data storage related to cardiac health, along with the aforementioned ideas, could significantly boost user engagement with the application, thereby enhancing CVD awareness.

7 Conclusions

This section provides an overview of the principal results achieved throughout the comprehensive system development process for CardioRef. CardioRef mobile android application was designed with the aim as an educational resource, assisting users in gaining a deeper understanding of cardiovascular diseases (CVDs) and the most effective strategies for their prevention and management. The following paragraphs provides a summary of the methods employed to achieve this aim through each specific objective.

- **Objective One (Background/Literature Review):** The background research conducted under the subtopics in Chapter Two; Background, established a robust framework for comprehending CVDs and their worldwide implications, as they represent the leading cause of death globally. Furthermore, this foundation research informed the project requirements and the needs of the target users as it includes a discussion on the various technologies and development methodologies. A fourth feature, the Quiz, was introduced to enhance user engagement within the app, drawing inspiration from the Background. Consequently, this influenced the attributes of the mobile environment, and the feature incorporated into the application development.
- **Objective Two (Animated videos on CVDs):** The CardioRef app incorporating animated videos related to CVDs on the pages dedicated to each type of CVD. These video, sourced from reputable providers, offers a more comprehensive understanding of the disease.
- **Objective Three (Dictionary on CVDs):** This application also features a dictionary that helps users in understanding and pronouncing common medical terms relating to CVDs, thereby enhancing their understanding of terminologies associated with CVDs
- **Objective Four (Newsletter or Articles on CVDs):** By integrating news and articles related to cardiovascular diseases (CVDs), each article or news headline directs users to the official website for further reading on the topic. This supplementary content will significantly enhance users' ability to discover information about CVDs, thereby aiding in their prevention and management.
- **Objective Five (Quiz on CVDs):** Based on the initial project description, project aim is to create an educational CVD Android application. However, after conducting research on the features of apps related to cardiovascular diseases (CVDs), I decided to add a quiz feature as a form of gamification. This was done by developing a quiz in a True/False format to assess users' knowledge about the disease.

The CardioRef application has made notable advancements in disseminating information regarding cardiovascular diseases (CVDs) and strategies for their management and prevention. This app represents a vital initiative aimed at enhancing understanding, prevention, and management of CVDs. By being developed on the Android operating system, it ensures accessibility for users, given that Android is the predominant platform for mobile devices. Nevertheless, it is evident that while the app effectively addresses many issues, there remain areas that require further enhancement, particularly in maintaining up-to-date information on CVDs and incorporating additional features to engage users more effectively. Despite these challenges, the app establishes a robust foundation for ongoing research and development related to CVDs and other non-communicable diseases.

7.1 Future Work

The CardioRef app presents significant opportunities for future development and enhancement. Prospective personalization features could include the capability to record vital signs, such as blood pressure, cholesterol levels, and pulse rates of users. Implementing this functionality would facilitate the monitoring of cardiovascular diseases (CVDs) and aid in their prevention or management. Furthermore, the incorporation of a physical activity tracking feature, combined with Artificial Intelligence, could enable the app to tailor activity plans for users based on their vital statistics and exercise habits.

Furthermore, a web-based version of this application could be created to function as a platform for the dissemination of free research theses and literature on cardiovascular diseases (CVDs), complementing the features offered by the mobile application. This platform would act as a central repository for the open exchange of knowledge regarding CVDs. Regarding future development, the application could be adapted for Apple iOS and broadened to include information on all non-communicable diseases.

Medical terminology within texts can be directly linked to a dictionary, enabling users to seamlessly transition between the textual information and the dictionary entries. This integration is likely to enhance the reading experience on the application, making it more engaging and enjoyable.

Future enhancements could encompass features such as the ability to save high scores, provide rewards for achieving the highest scores, and incorporate a greater variety of questions. These questions could be categorized by difficulty levels and may include quizzes with short answer

formats or multiple-choice questions (MCQs), along with the option to randomize the order of questions.

In summary, the potential for CardioRef to expand is vast, with opportunities to delve deeper into CVDs awareness, enhance user experience, and contribute significantly to public health and inclusion of all non-communicable disease. These future directions not only extend the capabilities of the CardioRef platform but also align with the aim of leveraging technology to foster education and awareness of CVDs.

In conclusion, CardioRef has significant potential for expansion, presenting opportunities to deepen awareness of cardiovascular diseases (CVDs), improve user experience, and make substantial contributions to public health with the inclusion of all non-communicable diseases. These prospective developments not only broaden the functionalities of the CardioRef app but also align with the objective of utilizing technology to promote education and awareness regarding CVDs.

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Appendix A Personal Reflection (Compulsory)

This **compulsory** appendix should contain a personal reflection on your project. It should contain two sections:

A.1 Reflection on Project

In reviewing the CardioRef project, I acknowledge the potential for enhancement in my project execution strategies. While I possess a strong foundation in general health and wellness, my understanding of cardiovascular disease (CVD) may not have included the most recent findings related to its prevention and management. For instance, I concentrated on conventional risk factors such as elevated cholesterol levels and smoking, while overlooking emerging studies that highlight the impact of mental health and air pollution on CVD risk.

Prior to the app's development, I conducted an extensive investigation into CVD prevention and the educational requirements of the intended audience. This involved reviewing numerous academic journals, examining Heart Association guidelines from various nations, and analyzing existing health applications to gain insights into their educational strategies. However, I did not dedicate sufficient time to exploring effective methods for engaging users through interactive content personalization, the storage of health vitals related to CVD, and the creation of tailored health routines based on these stored vitals, all of which could have significantly improved user experience and retention.

A significant challenge I encountered was making certain that the content was accessible to a diverse audience, particularly individuals with limited health literacy. Although the information provided by the app was medically sound, I observed that some users had difficulty understanding some of the technical terminology associated with heart disease. Furthermore, the app's design should incorporate usability enhancements to make app more user friendly.

Reflecting on the CardioRef project, I take pride in my achievements; however, these reflections also reveal areas for enhancement. I could have:

- Dedicated additional time to user testing with a varied demographic, particularly individuals who may lack familiarity with medical terminology or technology. This would ensure that the application is user-friendly and comprehensible.
- Enhanced content clarity by utilizing simpler language and incorporating visual aids such as infographics and diagrams to facilitate better understanding of the medical information.

- Placed greater emphasis on interactive features, such as gamification elements (e.g., progress trackers and challenges), to boost user engagement and retention. Additionally, I could have implemented reminders or tips to encourage users to interact with the app consistently.
- Established a feedback mechanism that allows users to pose questions or provide suggestions regarding the app, thereby enabling continuous improvement based on their input.

A.2 Personal Reflection

Upon reflecting on the development journey of the CardioRef project, I have gained valuable insights into both the functional and tactical aspects of app development. This experience has been highly educational, offering a thorough understanding of software development and its related processes.

If given the opportunity to start over, I would prioritize conducting user testing earlier in the development phase. I would seek feedback from a broader spectrum of potential users, including individuals from different age groups and varying levels of health literacy, to guarantee that the app is intuitive and accessible to all. Furthermore, I would implement more user-centered design principles, focusing on straightforward navigation and clearer instructions to enhance users' comprehension of the health information and tools offered by the app.

In the context of the content, I would enhance its interactivity by incorporating more engaging formats, such as gamified elements, to maintain user interest and promote the retention of educational material regarding the prevention and treatment of cardiovascular disease (CVD). Additionally, I would customize the content to cater to specific demographics, such as various age groups and individuals with pre-existing heart conditions, to ensure that the information presented is both relevant and captivating for the intended audience.

Ultimately, I will incorporate functionalities such as push notifications and progress tracking, enabling users to oversee their health objectives, including reducing cholesterol or managing blood pressure, directly through the application. Additionally, I will introduce a feature that permits users to share their progress with healthcare providers, fostering greater engagement between users and their physicians.

The project proved to be a well-rounded experience, combining both achievements and valuable lessons. The fulfillment derived from creating a functional Android application that enhances awareness of cardiovascular diseases (CVDs) and their health implications was significant. This endeavor deepened my knowledge of the Java programming language for app development, as well as my skills in using Figma for prototype design and applying scrum methodology in project management. These insights will serve as a foundation for my future pursuits, enabling me to create solutions that are both technically robust and user-centric.

Appendix B Ethics Documentation (Compulsory)

B.1 Ethics Pro-forma

Project Details

A1 Proposed Project Title

CVD Android Reference App (CardioRef)

Applicant Details

A2 Applicant Details

Title First Name Surname

Mr

Maxwell

Opore

Brunel Email address

2262365@brunel.ac.uk

A3 Are there other researcher(s) who will work on the research project?

☐ Yes

☒ No

A5 Applicant Status: Please select the capacity in which you are carrying out the research:

Undergraduate Student

A5.1 Please select your College:

College of Engineering, Design and Physical Sciences

Please select your Department:

Computer Science

Student Details

A7 Student Number

2262365

A8 Module Name and Number

CS3072-CS3605 Computer Science-Business Computing Final Year Project (2024/5)

A9 Assignment/Dissertation Submission Deadline Date

11/04/2025

A10 Supervisor Details

Title	First Name	Surname
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Dr

Nura

Abubakar

College

College of Engineering, Design and Physical sciences

Department

Computer science

Brunel email address

nura.abubakar@brunel.ac.uk

Health Research

A11 Does your research involve the recruitment of NHS patients and/or staff, HM Prison and Probation Services, or require approval from the Social Care Research Ethics Committee?

☐ Yes

☐ No

Risk Factors

A12 Does the research fit into any of the following security-sensitive categories?

- ☐ Concerns the military
- ☐ Set up under an EU security call
- ☐ Involves seeking a security clearance
- ☐ Concerns terrorist or extremist groups

☒ None of the above

A13 Does the research involve the use of human participants, their data and/or their tissue? Please tick yes if your research includes data obtained through questionnaires/surveys, face-face/phone interviews, focus groups or sensitive financial data.
See the information (i) icon for guidance.

- ☐ Yes
☒ No

A13.1 Are the data you intend to use open source or closed source? Please see the Information Icon for more information.

Open Source

A13.1.2 Does your research involve use of social media or online forums?

- ☐ Yes
☒ No

A13.1.6 Please provide details of your open data sources.

CVD related articles from google scholar, World Health Organization and other relevant medical websites.

Please give details of your proposed methodology and your project aims.

The software development will use agile methodology to create an educative and awareness mobile android app on CVD.

Research Integrity

A23 Have you completed the Research Integrity Online Training relevant to your field of research (see the information icon for instructions)?

- ☒ Yes
☐ No

A23.1 If yes, please provide details:

I did have read on the ethical research since that is related to the app am building

Start and End Date

J1 Proposed Start Date

03/12/2024

NOTE: You are not able to select a start date within a week of your application. If you are using human participants, please select a suitable start date that allows time for the review and approval of your application. **Please see the information icon (i) for further guidance.**

J2 Proposed End Date:

10/12/2024

Researcher/Applicant

Researcher/Applicant Declaration

- ☒ I understand that I cannot commence my research until full and final approval has been granted by the relevant Research Ethics Committee.
- ☒ I understand that it can take up to 15 working days to receive feedback on my application.
- ☒ I understand that once approval has been granted, I may conduct my study only as set out in my approved application, and in accordance with any conditions set out in the approval letter.
- ☒ I confirm that the research will be undertaken in accordance with the following;

[Brunel University London Code of Research Ethics](#), and [Brunel University London Research Integrity Code](#).

- ☒ I shall ensure that any changes in approved research protocols are reported promptly for approval by the relevant Research Ethics Committee.
- ☒ I shall ensure that the research study complies with the law and Brunel University London policies on the use of human material (if applicable) and health and safety.
- ☒ I am satisfied that the research study is compliant with the Data Protection Act 2018, and that necessary arrangements will be made with regard to the storage and processing of participants' personal information and generally, to ensure confidentiality of such data supplied and generated in the course of the research.

(Note: Where relevant, further advice is available from the University Data Protection Team, e-mail data-protection@brunel.ac.uk).

- ☒ I will ensure that all adverse or unforeseen problems arising from the research project are reported in a timely fashion to the Chair of the relevant Research Ethics Committee.
- ☒ (For members of staff and PhD students) I will undertake to provide notification to the Chair of the relevant Research Ethics Committee when the study is complete, or if it fails to start or is abandoned.

Researcher/Applicant Signature:

Signed: This form was signed by Maxwell Opare (2262365@brunel.ac.uk) on 26/11/2024 11:00

Supervisor

Signature of Supervisor

- I confirm that I have advised the student on the ethical aspects of the study design and their responsibilities in relation to the submission of this application and the research.
- I confirm that the student has been advised to read the University's Code of Research Ethics and other relevant documentation.
- I confirm that the student has the skills to carry out the research.
- I confirm that if the research involves human participants, a Participant Information Sheet (PIS) is included and has been completed appropriately.
- I confirm that if the research involves human participants the procedures for recruitment and obtaining informed consent are appropriate.
- I confirm that if there are issues of risk in the research a full risk assessment has been undertaken and is attached.
- I confirm that a DBS check has been obtained (where appropriate).

Signed: This form was signed by Nura Abubakar (Nura.Abubakar2@brunel.ac.uk) on 26/11/2024 14:54

Appendix C Video Demonstration (Compulsory)

C.1 Link to the video

Video link: <https://youtu.be/8Nayhb7pwSM>

Figma prototype link:

<https://www.figma.com/proto/6yK1zQd7hCCt6f9e7KwQAh/CardioRef?node-id=0-1&t=fFdMDv4OwbMVwED9-1>

Appendix D Other Appendices

More relevant material

The remaining appendices can contain relevant material which is not essential to be included in the main body of the dissertation, but which may be useful to support your dissertation.

- Examples of relevant material might include:
- Example questionnaires
- More detailed designs
- Relevant results which didn't fit in the main body

Examples of material that should not go into an appendix:

- A dump of all your code
- Transcripts of all your interviews

Remember, that the appendices should be there in case the reader wants to refer to them. They will not be read as part of the dissertation story, so do not just use them to put essential material because you ran out of room. Also, do not be tempted to pad out your dissertation to the full 60 pages just by adding lots of unnecessary material to the appendices.

Any supplementary materials can be uploaded electronically with your submission.